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Review

Arabic Natural Language Processing (NLP): A Comprehensive Review of Challenges, Techniques, and Emerging Trends

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Abstract

Arabic natural language processing (NLP) has garnered significant attention in recent years due to the growing demand for automated text and Arabic-based intelligent systems, in addition to digital transformation in the Arab world. However, the unique linguistic characteristics of Arabic, including its rich morphology, diverse dialects, and complex syntax, pose significant challenges to NLP researchers. This paper provides a comprehensive review of the main linguistic challenges inherent in Arabic NLP, such as morphological complexity, diacritics and orthography issues, ambiguity, and dataset limitations. Furthermore, it surveys the major computational techniques employed in tokenisation and normalisation, named entity recognition, part-of-speech tagging, sentiment analysis, text classification, summarisation, question answering, and machine translation. In addition, it discusses the rapid rise of large language models and their transformative impact on Arabic NLP.

Keywords: Arabic NLP; Arabic LLMs; Arabic NLP challenges

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1. Introduction

Arabic natural language processing (NLP) has engendered remarkable progress in recent years, enhanced through innovations in deep learning, transfer learning, and large language models (LLMs). Arabic is one of the most widely spoken languages in the world, estimated to have over 400 million speakers [1]. It constitutes a linguistically and culturally significant medium for global communication, education, and information access. Despite its global impact, Arabic is comparatively under-observed in computational linguistics research, largely due to its complex orthography, rich morphology, and ambiguity and polysemy, as well as the limited availability of high-quality resources and the variability between modern standard Arabic (MSA) and regional dialects. MSA is formally utilised across educational, journalistic, and official domains, whereas dialects informally dominate in social media content. The widespread dissemination of digital Arabic language content across news, social media, and educational contexts has generated significant opportunities and challenges within the field of NLP. NLP in general and Arabic NLP in particular have various subfields, such as text classification, text summarisation, and machine translation. A variety of approaches have contributed to NLP fields, starting with traditional methods (e.g., rule-based parsing and statistical learning). These were followed by machine learning and deep learning algorithms, which have significantly advanced NLP. Finally, transformers, LLMs, and pretrained model architectures have dramatically improved machine understanding. However, significant challenges continue to hinder the

development of reliable standardised orthographic normalisation, tokenisation, and comprehensive datasets that capture the linguistic diversity of Arabic.

This paper aims to provide a comprehensive review of the state of Arabic NLP tasks. The review begins by analysing and distinguishing the key linguistic and technical challenges impacting the effective processing of Arabic text. It details the related issues in this context, including complex morphology, complicated orthography and diacritics, ambiguity of significance in context and words, and dataset scarcity. Following this, it explores the principal computational techniques, models, approaches, datasets, and evaluation methods used in various Arabic NLP tasks, such as tokenisation, named entity recognition (NER), sentiment analysis, text classification, summarisation, question answering, and machine translation. Furthermore, it surveys the growing influence of LLMs and transformer-based architectures that have redefined Arabic NLP research in recent years. Finally, this paper discusses emerging trends, current limitations, and future research directions, emphasising the need for inclusive datasets, dialectal coverage, and unified evaluation standards to advance the development of robust, sustainable, and equitable Arabic NLP systems.

2. Methodology

The major objective of this paper is to comprehensively review the existing studies on Arabic NLP, with the ultimate aim of highlighting the challenges, techniques, and emerging trends addressed in previous studies. This paper also aims to show the most effective techniques and approaches applied across various Arabic NLP fields. Also, it offers insights into potential methodologies to enhance future Arabic NLP research and applications. The review was conducted in the period between January 2025 and October 2025. The present comprehensive review followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [2]. While undertaking this Arabic NLP literature review, it took the following steps:

2.1. Determining the Research Questions

The review mainly focuses on Arabic NLP based on textual data and therefore in this study, the research questions are as follows:

1. RQ1: What are the main challenges and limitations in Arabic NLP for different fields?
2. RQ2: What methods, tools, and models have been commonly applied in each Arabic NLP task?
3. RQ3: What publicly available datasets exist for each Arabic NLP field?
4. RQ4: What evaluation techniques have been applied to measure the performance and effectiveness of each trending methodologies?

2.2. Search Strategy

This comprehensive literature yielded over 300 references, that focused on recent and emerging studies in Arabic NLP. Over 60% of the studies reviewed in this paper were published within the past decade, reflecting the recent surge of research activity in Arabic NLP. Furthermore, the origins of research interest in this field can be traced to the early years of the 21st century, during which several foundational tools and preliminary techniques were developed. Publications from this period constitute approximately 17% of the total studies reviewed in this paper. Moreover, a limited number of earlier studies, published prior to 2000, addressed foundational issues and challenges associated with Arabic language processing. As illustrated in Figure 1, the distribution of references demonstrates a notable increase in Arabic NLP research output, particularly during the period from 2020 to 2025, reflecting the rapid expansion and growing academic interest in the field.

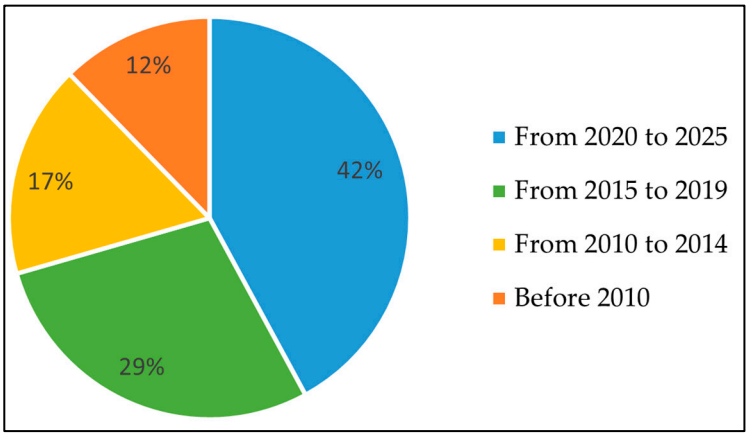

Figure 1. Distribution of references over the years.

2.3. Article Selection

In addition, the review encompassed leading electronic databases, including ACL Anthology, ScienceDirect, SpringerLink, IEEE Xplore, ACM Digital Library, arXiv, and MDPI, which collectively comprise the primary sources of references utilised in this study. Along with supplementary sources such as Google Scholar and ResearchGate. Also, other supplementary sources such as Google Scholar and ResearchGate were prioritised highly cited and influential studies to ensure the inclusion of foundational and representative research in the field of Arabic NLP. Figure 2 shows the correlation between the source of electronic databases and the cited number for each database. The review incorporated diverse types of references, such as research article, proceedings/conference article, books, datasets, or technical report, to ensure comprehensive coverage of the field. Studies were excluded if they were not related to textual Arabic NLP tasks, not written in English, or lacked full-text access.

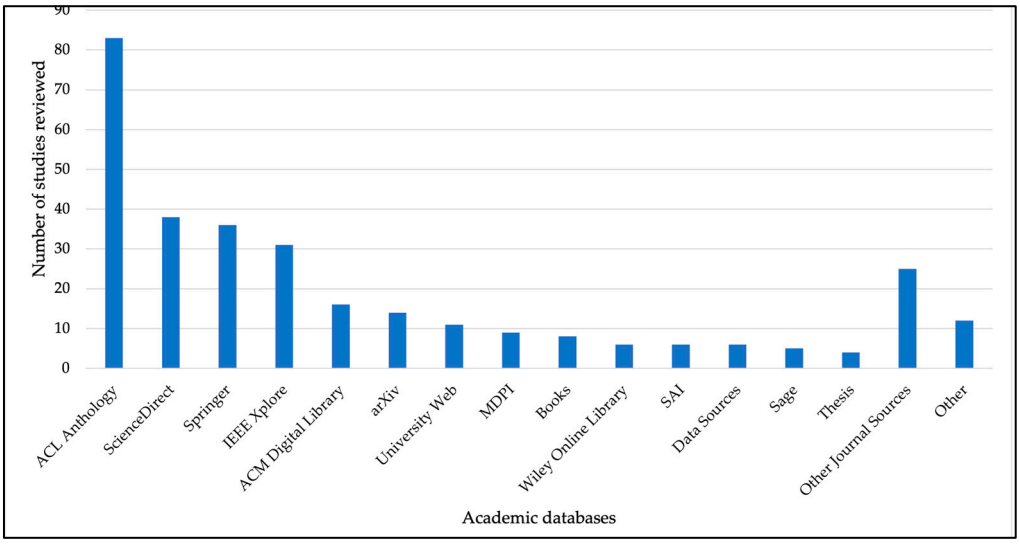

Figure 2. Distribution of relevant studies retrieved from each electronic database used in the Arabic NLP review.

2.4. Validation

All studies relevant to Arabic NLP were retrieved and managed systematically. The search results from each database were exported in BibTeX format and organized using the Zotero reference management tool. Each publication was thoroughly reviewed to evaluate its quality and relevance, ensuring that the selection process maintained a high level of objectivity, transparency, and reproducibility of results.

3. Challenges in Arabic NLP

The Arabic language poses various challenges to NLP research fields, with these issues arising from the unique nature of Arabic in terms of its linguistic characteristics and complexities. NLP in both Arabic and English shares common challenges, including ambiguity, polysemy, and the need for extensive annotated datasets. Also, context-dependent meanings and semantic variations are required to achieve accurate natural language understanding. Yet, Arabic presents distinct challenges due to its specific linguistic features. One of the differences lies in the richness of Arabic morphology, characterised by complex word structures, affixes, and templatic patterns, whereas English has relatively simpler morphology. Additionally, the considerable diglossic differences between MSA and regional dialects contrast with the more standardised usage of English across regions. Moreover, the use of diacritics to represent short vowels affects pronunciation and meaning, leading to orthographic variations that are far less pronounced in English.

These challenges include the morphological richness of the language, which results in Arabic words having considerable variety in derivation and inflection. Non-native speakers may face considerable ambiguity due to the absence of diacritical marks, as the lack of these marks causes the meaning of many words to change based on the context even as their spellings stay the same. Furthermore, there are numerous dialects in Arabic due to the expansive geographical distribution of the Arab countries as well as the large number of dialects in each country. In addition, variations in word order and sentence structure further contribute to difficulties in processing Arabic text, particularly due to the lack of standardisation across different contexts. Moreover, one of the main challenges is the scarcity of labelled datasets and other linguistic resources, especially those focused on dialectal Arabic. This shortage hampers the development of reliable, accurate, and powerful NLP tools. These challenges can be addressed by developing approaches and resources tailored to the complexities of Arabic language processing. The following subsections outline some of the key challenges in Arabic NLP.

3.1. Complex Morphology

Arabic is considered one of the most morphologically intricate languages in the world, and its complexity arises from multiple factors. One is the root-and-pattern morphological framework, which consists of a root of consonants with an abstract vowel, consonant, or both patterns [3]. The root of Arabic words usually comprises three consonants, representing the core semantic concept of the word [4]. For instance, the root of the Arabic word 'كتب' is k-t-b, which is related to the concept of 'writing'. It yields various forms by combining in different ways (e.g., 'كتاب' *kitāb* ('book'), 'كَتَبَ' *kataba* ('he wrote'), 'مكتب' *maktab* ('office'), and 'مكتوب' *maktūb* ('written')) [5]. Figure 3 illustrates Arabic words derived from the root كتب (k-t-b). The arrows on the left indicate the morphological relationships among these words, showing how prefixes and vowel changes generate new meanings while retaining the original root. This allows Arabic to express a variety of related meanings efficiently and the non-linear construction of the derived words occurs by integrating consonants and vowels [6]. This structure applies specific templates on the consonantal root, thereby creating diverse nouns, verbs, and adjectives with semantic and grammatical coherence as well as versatile expression [7]. Moreover, Arabic words can be formed in various formats due to the word's inflection, its derivation, and a variety of morphological features. A single root or derived word in Arabic can be constructed in many forms by attaching various prefixes, suffixes, and infixes to it [8]. Such an affix can indicate various grammatical and semantic features, such as gender (masculine and feminine), number (singular, dual, and plural), possession, and verb conjugations [9]. This is called derivational morphology in Arabic, and it is highly conducive to word formation from a single root [10].

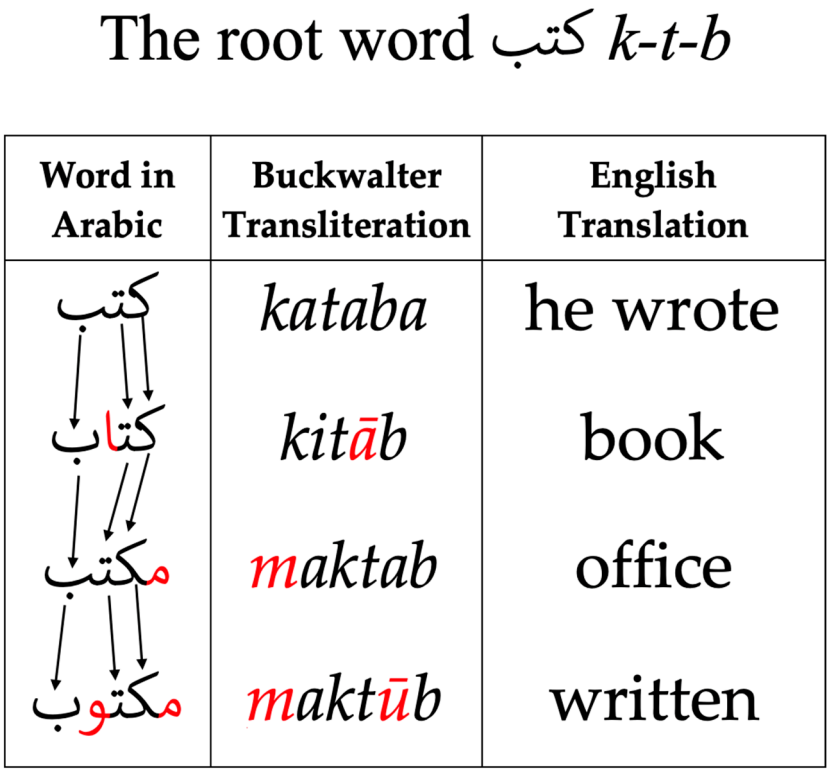

Figure 3. Examples of derived words from the Arabic root (k-t-b), showing their Buckwalter transliteration and English meanings. The arrows illustrate how different words are derived from the same root. The red letters indicate the varying positions of the root consonants (**k–t–b**) within each derived word.

3.2. Diacritics and Orthography

Arabic has only three vowels litters (ا ، و ، ي), and they are all long vowels. There are also short vowels and other phonetic instructional support for comprehensive reading in Arabic, and they are called diacritics There are also short vowels and other phonetic instructional support for comprehensive reading in Arabic, and they are called diacritical [11]. These are small diacritical marks positioned above and below Arabic characters [12]. In addition, Arabic utilises diacritics to modify the pronunciation and meaning of words [13]. However, their absence in standard writing poses challenges for computational analysis and non-native language acquisition [14]. For example, the word 'كتب' *ktb* without diacritics can have different meanings based on contextual and syntactic clues. It can be represented with short vowels as 'كَتَبَ' *kataba* ('he wrote'), 'كُتُب' *kutub* ('books'), and 'كُتِبَ' *kutiba* ('was written') [15]. These diacritical marks are common in formal MSA texts. Overall, they can lead to equivocation, especially with long vowels in the orthography [16].

Furthermore, orthography in Arabic is a complicated task for multiple reasons. First, the phonetics of short and long vowels in Arabic may lead to confusion due to their similarity, especially on social media platforms [17]. Second, Arabic has many forms for some letters, such as 'hamza letter' (ء ، ئ ، ؤ ، آ ، ا ، إ ، أ), and their phonetics are very similar; similarly, both 'taa marbuta' and 'ha' letters at the end of a word have similar phonetics. Many researchers tend to normalise all different forms into one letter [18]. Another reason is that the translation or transliteration of foreign named entities into Arabic can spell the words contradictorily based on their phonetics [19]. The numerous spoken dialects in the Arab world and Arabic being the most written language on social media comprise another orthography challenge [20]. On social media platforms, the variations of Arabic dialects follow no standard orthographic conventions, instead often relying on colloquial spellings and phonetic changes. However, MSA relies on a fixed and uniform orthographic rule [21].

3.3. Ambiguity and Polysemy

In Arabic, the context can affect the meaning of a single word; thus, Arabic words often have various meanings. Therefore, the terms 'polysemy' and 'ambiguity' are used to distinguish features of the Arabic language, each capturing its complexity and richness [22]. Both share common characteristics but are differentiated slightly. Ambiguity occurs when a single word has multiple distinct wider ranges of interpretation. For instance, the word 'عين' in Buckwalter transliteration is '*Eyn*', and it can hold different meanings, which are 'eye', 'spring', or 'spy' depending on the context of the usage [23]. In contrast, polysemy occurs when a single word has multiple linked interpretations of the meaning of the word. This often derives from its root-based morphology in Arabic. For example, the word 'عقد' in Buckwalter transliteration is '*Eaqd*', and it means 'to tie' or 'to knot'. However, it can have extended meanings, such as 'contract' or 'agreement' in 'الزواج عقد', which means 'marriage contract'; 'to hold' or 'convene' in 'اجتماع عقد', which means 'to hold a meeting'; and 'to organise' or 'arrange' in 'مؤتمر عقد', which means 'to organise a conference'. Different examples have related meanings or ideas that are brought together. Both of these traits emphasise the eloquence of Arabic, allowing for various meanings in the literature and speech [24].

3.4. Challenges with Arabic NLP Datasets

Arabic NLP datasets are the most common resources used for applying different NLP tasks and building various NLP tools. There are many available datasets for NLP purposes, but there remains a need for large, high-quality datasets. The main limitation is that existing datasets are primarily focused on MSA or a few specific dialects, whereas datasets covering other dialects are scarce. Each dialect is distinct, with its own style in terms of phonetics, vocabularies, syntax, and structures [25]. Moreover, there is a paucity of annotated and labelled Arabic NLP datasets and a lack of accurate annotated tools and guidelines [26]. Furthermore, most existing Arabic NLP datasets are limited in quality due to the absence of diacritics or the presence of data noise, particularly those retrieved from the web [27]. The available datasets also concentrate on several specific domains that yield formal texts, such as politics, commerce, and other news data. This renders them inappropriate for informal or other domain-specific tasks in Arabic NLP [28]. In addition, the variation in the web resources for the collected datasets and available corpora leads to noisy data without standardisation as well as overlapping text data [29]. Moreover, labelling and annotating inconsistencies across Arabic NLP datasets result in complications in terms of integrating resources effectively [30]. Overcoming these challenges necessitates collaborative efforts to propose a large balanced and annotated corpora for MSA and different dialects, with a standardisation process.

4. Techniques in Arabic NLP

There are a range of supportive methods that seek to handle the unrivalled complexities of the Arabic language. They revolve around simplifying the rich morphology using tokenisation, stemming, lemmatisation, or segmentation techniques, which reduce the many forms to the roots and base forms, or around splitting the root from the affixes. This is a very crucial task due to the nature and structure of Arabic words, especially the root-based ones, which can appear in many forms in text. In addition, addressing flexible syntax includes part-of-speech (POS) tagging and NER. Both can aid in recognising the structure of Arabic sentences and increase the machine's ability to understand Arabic text for different NLP tasks. Moreover, there are other techniques that can minimise the various forms of a word with regard to the different forms of a letter using text normalisation techniques or eliminate the presence of short vowels using diacritic restoration techniques. Both tech-

niques can improve the machine's efficiency in handling different Arabic NLP tasks. The use of machine learning models, particularly deep learning and transformer-based models, has engendered remarkable development in a variety of Arabic NLP tasks. They offer a fine-tuned process for Arabic NLP tasks, overcoming barriers in machine translation, sentiment analysis, and language generation. The techniques in Arabic NLP can be grouped as follows.

4.1. Text Tokenisation and Normalisation

The nature of Arabic words can be classified into three types: nouns, verbs, and particles [31]. Nouns and verbs are the most diverse in their forms due to the non-limitation of affixation forms [32]. In Arabic, affixations are embedded within words, and they can occur before the root, in the middle of the root, or after the root [33]. Therefore, tokenisation, which breaks the text into meaningful units (usually words), is very challenging because of irregularities in deriving forms of Arabic words [34].

Arabic text normalisation is the process of unifying the redundancy of the various forms of a word [35]. As previously stated, Arabic has a rich morphology and uses diacritics, and some Arabic character shapes are diverse [36]. This high variation in Arabic text makes text normalisation essential for ensuring performance consistency for different NLP tasks [37]. Moreover, informal Arabic is the form primarily used in the casual communication that occurs between users of social media and other non-formal platforms [38]. Some common issues that may occur as a result are extra spaces, spelling errors, elongated duplicate characters (especially vowel letters for emphasis), and various dialects. This adds further intricacy to the normalisation process. Nevertheless, standardising Arabic text normalisation helps improve the machine's ability to conduct critical tasks, such as machine translation, sentiment analysis, and text classification [39]. This step is crucial for reducing the noise resulting from the multiple shapes of Arabic words, and it can efficiently deal with the intricacies of the Arabic language in different contexts and applications. The three techniques most commonly applied for normalising Arabic words are stemming, lemmatisation [40], and segmentation [41].

There are many tools and algorithms for word stemming, word lemmatisation, and word segmentation in Arabic that can effectively represent the features of Arabic words. Word stemming in Arabic NLP is the process of converting Arabic words into their root forms by eliminating the affixes from a word [8]. There are many Arabic stemming algorithms. Khoja stemmer [42] is one of the earliest and most well-known Arabic stemming algorithms. It is based on a predetermined list of Arabic roots as well as pattern-based rules for identifying roots. It analyses Arabic word morphology but has a shortage of out-of-root words and other irregularly inflected forms [43]. Larkey's light stemmer [44], proposed by Larkey, Ballesteros, and Connell, tends to apply a rule-based approach. It eliminates affixes from words; however, as it does not consider whether the right affixes have been removed, it does not necessarily return a valid form of the root. It is applied for information retrieval tasks. The information science research institute (ISRI)stemmer [45] is a light stemmer tool that is similar to Khoja stemmer [42], but it lacks a predefined list of roots. This tool removes common affixes without strictly implementing root extraction. Enhanced algorithm for Arabic stemmer [46] introduced an enhanced root-based algorithm that trims all affixes, including prefixes, suffixes, and infixes, based on morphological patterns. Light and heavy Arabic stemmer [8] has three processing phases for root extraction: eliminating Arabic affixes (prefixes and suffixes), identifying Arabic verb patterns, and evaluating the root form. P-Stemmer [47], introduced by Kanan and Fox, is an altered form of Larkey's light stemmer. It eliminates only the prefixes of a word while conserving a better word output. The main goals are avoiding excessive truncation and preserving the

accurate meaning of the word compared to other stemmers that remove both prefixes and suffixes. Tashaphyne 0.4 stemmer [48] is based on the Rhyzome model, and it has three main stages to extract Arabic 'roots' and 'stems' from Arabic text. They are preparation (tokenisation), stem-extractor (using a modified finite state automaton), and root-extractor.

Word lemmatisation is another NLP normalisation technique, and it entails converting Arabic words into their dictionary or base form (lemma). In contrast to stemming, which removes affixes without considering the correct grammatical form of the output word, lemmatisation produces a meaningful output word form based on linguistic rules. Several Arabic lemmatisation tools are available. MADA + TOKAN [49] is a toolkit that provides various methods for extracting contextual and morphological information from Arabic text, and one of these methods is lemmatisation. It has two main parts: MADA and TOKAN. MADA uses support vector machine (SVM) models to choose the best match for the analytical meaning of the current word from a list of options. Then, TOKAN employs the output of MADA to produce the tokenised output. Arabic lemmatizer [50] implements hidden Markov models (HMMs) [51] to nominate the proper lemma result from all possible generated options during the morphological analysis. It can process text with and without diacritics. AlKhalil Morpho Sys [52,53] proposed two versions of a morphological analyser tool for standard Arabic. The first version deals with words with full or partial diacritics. The second version is more advanced in terms of performance accuracy and database enrichment, in addition to possessing new features, including lemmatisation. Alma [54] was built based on an ordered frequency dictionary that included words with and without diacritics. It was obtained from the Qabas lexicographic database [55], and it is a remarkable tool for producing unambiguous lemmas. Arabic Treebank (ATB) [56] and the Salma corpus [57] were used to evaluate the tools and compare the results with other models.

Word segmentation in Arabic NLP is the process of splitting Arabic words into the following parts: prefix, root, and suffix [58]. It is unlike stemming and lemmatisation techniques in that it keeps the affixes of a word as separable units from the stem or the root. Many Arabic segmentation tools and techniques exist. Morpho-syntactic [59] introduced some Arabic word segmentation techniques, namely supervised learning (SL), the frequency-based approach (FB), finite state automaton-based approach (FSA), and improved FSA (IFSA). IFSA has three advantages: consistent enhancement of baseline performance across tasks, efficient implementation in a large corpus, and flexibility in a diversity of tasks. Integration of a segmentation [60] was built based on the NooJ platform [61] using a set of punctuation-based rules as well as NooJ transducers using lexical-based rules to reduce parsing complexity. The linguistic and graphic segmentation approach [62] offers a segmentation technique that uses linguistic and graphical connectors as well as initial algorithm prototypes. Clitic segmentation [63] is a unified segmentation model for both formal and dialectal Arabic texts. It utilises dialect-independent features and simple domain adaptation. SVM-based and Bi-LSTM-CRF segmentation [64] offers a segmentation approach for four Arabic dialects: Gulf, Egyptian, Maghrebi, and Levantine. It uses two segmentation techniques, which are ranking based on SVM and sequence labelling using Bi-LSTM-CRF. DJAZI segmentation [65] is a tool for Arabic text segmentation that combines two approaches: contextual exploration at the level of the text and morphological segmentation at the level of the word. Table 1 presents a comparison of all the mentioned Arabic NLP tools, outlining their main approaches, and typical applications across stemming, lemmatisation, and segmentation tasks.

Table 1. Comparison of key Arabic NLP tools by approach and normalisation technique.

Normalisation Technique	Tool	Type/Approach
stemming	Khoja stemmer [42]	List of Root-based and Pattern based
	Larkey's light stemmer [44]	Simplified root-based
	(ISRI)stemmer [45]	Light common affix stripping
	Enhanced algorithm for Arabic stemmer [46]	Enhanced root-based algorithm affix stripping
	Light and heavy Arabic stemmer [8]	Enhanced light and heavy root-based algorithm
	P-Stemmer [47]	Light prefixes only stemmer
	Tashaphyne 0.4 stemmer [48]	light stemming algorithm based on the Rhyzome model
lemmatisation	MADA + TOKAN [49]	Rule-based morphological analyzer
	AlKhalil Morpho Sys [52,53]	Extensive morphological rules and linguistic datasets
	Alma [54]	A frequency-based morphological dictionary and Qabas lexicographic database [55]
segmentation	Morpho-syntactic [59]	Hybrid supervised learning, frequency-based, and finite-state automaton approaches
	Integration of a segmentation [60]	Based on punctuation signs extracted from a study corpus
	The linguistic and graphic segmentation approach [62]	Based on linguistic and graphic connectors
	SVM-based and Bi-LSTM-CRF segmentation [64]	SVM ranking and Bi-LSTM-CRF sequence labeling
	DJAZI segmentation [65]	Hybrid contextual text exploration with word-level morphological segmentation

Several tools have been developed to deal with the complexities of Arabic language processing. They have several functionalities, such as segmentation, lemmatisation, stemming, spellchecker, POS tagging, discretisation, and NER, along with dialect identification and other special functions. There are widely used tools with outstanding evaluation results that provide most of these features. For example, MADAMIRA [66] is an advanced and enhanced integrated model from previous tools, namely MADA [49] and AMIRA [67]. It uses SVMs and n-gram language models based on its morphological analysis pipeline to apply Arabic morphological features and analyse Arabic text [66]. Farasa utilises SVM with linear kernels to rank for segmentation purposes, with faster implementation in terms of machine translation and information retrieval [68]. Furthermore, Farasa uses SVM-rank with linear kernels along with lexical and morphological features for segmentation [69]. CAMeL tools employ deep learning models and support MSA and dialectal varieties [70]. Stanford CoreNLP is a lightweight annotation-driven NLP toolkit, and it can easily be adapted alongside tools such as NLTK [71].

4.2. Named Entity Recognition (NER)

NER is an NLP-related process that has the capability to identify and recognise named entities from unstructured text and then classify them into categories [72]. The algorithms can automatically detect named entities, extract their information, and categorise them into key subjects, including personal names, locations, organisations, events, and dates [73]. There are many studies striving to improve the field of NER for Arabic. CANERCorpus is the classical Arabic named entity recognition corpus, which is annotated with unique named entity classes for Islamic topics derived from over 7000 hadiths [74]. MADAMIRA's Arabic NER can correctly identify named entities based on its morphological analysis pipeline [66]. Farasa NER is an accurate Arabic NER that utilises machine learning techniques to recognise entities within text, and it is part of the Farasa NLP toolkit [68]. CAMeL NER, one of the functionalities in CAMeL tools, employs deep learning models and supports MSA and dialectal varieties [70]. The author of [75] built an Arabic NER model that is a combination of three layers: a transformer-based language model layer, a fully connected layer, and a conditional random field (CRF) layer. Furthermore, a proposed hybrid Arabic NER technique integrating rule-based and machine learning ML in a pipelined process has been found to outperform 11 standalone entity types with good achievement results [76]. The Tafsir dataset is a multi-task benchmark for Arabic NER and topic modelling

purposes that was built manually with over 51,000 annotated sentences [77]. In addition, a novel Arabic NER framework that effectively handles complex and overlapping entities has been introduced; it uses advanced architectural components, namely hybrid feature fusion, compound span representation, and enhanced multilabel classification [78]. NER was utilised with multi-task learning (MTL) as a feature to enhance deep learning models for detecting Arabic fake news [79].

The benefits of NER can positively impact many NLP domains. For example, NER can support machine translation tasks [80] and automatically extract information for retrieval purposes [81]. Furthermore, NER improves text clustering [82]. Question answering is another field that has been positively affected by NER [83], and text summarisation shows promising results when using NER [84]. A challenge in Arabic NER is the absence of capitalisation; unlike in English, there are no uppercase and lowercase letters to distinguish the beginning of names in Arabic. Other issues include the agglutination of affixes to Arabic words [85], the misspellings of some Arabic words [86], short vowels [87], and ambiguity in the meaning of the same word [88]. These challenges can arise from either the scarcity of high-quality annotated datasets or the complexities of entity forms. Various Arabic NER tools have been developed. For instance, NooJ enables rule-based NER system design using finite-state and context-free grammar rules [61]. The Buckwalter Arabic Morphological Analyzer (BAMA) provides rich lexical resources, morphological analysis, and transliteration support for improved readability and entity disambiguation [89]. Moreover, the AMIRA tool includes a tokeniser and a POS tagger, and it has been widely used in various applications, especially Arabic NER research [67]. MATAR is an open-source tagger for Arabic that supports both automatic and manual tagging using general or customised morphological tags [90].

4.3. Part-of-Speech (POS) Tagging

POS tagging is the process of indicating the grammatical information of the words in a sentence based on their definition and their occurrence within the context [91]. It simply classifies words into categories, such as nouns, verbs, adjectives, and adverbs [92]. It is useful for capturing the syntactic and semantic details of words to understand the structure and meaning of a sentence [93]. This technique benefits several NLP tasks, such as machine translation, NER, grammar checking, and information retrieval [90]. There are many tools that can perform Arabic POS tagging, with MADAMIRA [66], Farasa [68]. CAMeL tools [70], and Stanford CoreNLP [71] being some of the widely used ones. Furthermore, Fassiehreg is an interactive Arabic annotation tool that performs high-accuracy Arabic morphological analysis, one of which is POS tagging. It combines statistical disambiguation with a user-friendly interface [94]. A POS tagging approach for Tunisian Arabic that leverages its similarity to MSA was presented through morphological analysis, lexical transfer, and morphological generation [95]. The Standard Arabic Profiling (SAP) toolset consists of POS, vocabulary, and readability profilers. It uses the Stanford CoreNLP tagger for POS analysis and a vocabulary profiler through comparison with the open source Arabic corpus (OSAC) corpus using log-likelihood measures [96]. The Arabic extended morphological analysis and disambiguation tagset (EMAD) is an intermediate tag set and a corresponding system for unifying various Arabic POS annotation schemes [97]. Tasaheel is an automated Arabic textual analysis tool that provides various features related to Arabic NLP tasks. In addition, it enhances traditional POS tagging by including detailed POS summaries as well as emotion- and domain-specific tagging, thereby offering in-depth linguistic insights not previously available in Arabic NLP tools [98]. Other tools include an integrated Arabic POS tagger based on first-order Markov and decision tree models that was trained on the network for euro Mediterranean language resources (NEMLAR)

corpus [99] and a bidirectional encoder representations from transformers (BERT)-based Arabic POS tagger trained on an integrated Arabic WordNet and the Quranic Arabic Corpus [100].

The author of [101] achieved a high F-measure for Arabic POS tagging using a statistical HMM-based approach, along with a 55-tag set and Buckwalter's stemmer. Moreover, Arabic POS tagging was presented using two phases: combining the Alkhalil Morpho Sys morphological analyser [52] with smoothing techniques and statistical analysis [102]. Three new manually annotated Arabic POS tagging datasets were presented, namely MSA, Gulf dialects, and mixed dialects. They were collected from X (formerly Twitter), and supervised CRF and Bi-LSTM models were applied [103]. A rule-based model using Arabic POS tag extraction techniques was introduced to identify subject–predicate–object triples [104]. The POS tagging technique has been used in many studies related to Arabic text classification, such as combining stemming and POS tagging techniques for text classification [105] and comparing different stemmers; a root extractor with a POS tagger showed the best performance [106]. In addition, POS tagging was used for real and fake news detection [107], and a POS tag algorithm was introduced for identifying narrators' names and sanad types in hadith texts [108]. There exists extensive research showing the impact of using POS tagging in Arabic sentiment analysis [109]. A study compared SVM-rank and Bi-LSTM for Arabic POS tagging, showing that SVM-rank achieves high accuracy through extensive feature engineering but that Bi-LSTM can automatically learn linguistic features [110].

4.4. Lexicon

A lexicon is an assembled linguistic resource that refers to a list of words and maps them to certain comprehensive features to enable machines to understand and disambiguate human language [111]. Alternatively, it can be described as a structured source that provides special interpretation, meaning, information, or grammatical properties for vocabulary [112]. Moreover, it can offer insights into dealing specifically with Arabic dialects alongside MSA. Due to the extensive complexity of the Arabic language, across various NLP tasks, lexicons have critical impacts that enable more accurate language modelling [113]. Lexicons in Arabic NLP are a fundamental resource for supporting a wide range of Arabic NLP tasks, such as morphological analysis, POS tagging, NER, and sentiment analysis.

An Arabic lexicon, namely the SALMA–ABCLexicon, was employed in [114] to improve the morphological analysis. ElixirFM is an online Arabic morphological analyser for MSA [115], built based on a version of the Buckwalter lexicon [116]. MAGEAD is another morphological analyser and generator for both MSA and its dialects, constructed by representing pairs based on Elixir-FM's extended lexicon [115] and applying detailed morphophonemic and orthographic rules. BAMA is one of the well-known analysers [117,118], and it is used for morphological analysis and POS tagging. It contains over 77,800 stem entries [116] and approximately 83,000 entries of Arabic prefixes, suffixes, and stems [117]. Furthermore, there exists a large-scale Arabic morphological analyser and generator presented using lexicons from other languages' transducers and rules adapted from two-level morphology (KIMMO-style system [118]) using Xerox tools [119]. An Arabic morphological analysis and generation tool was introduced based on a reduced lexicon of the Arabic root-and-pattern structure [120]. Alma [54] is an open-source Arabic lemmatiser, as well as POS and root tagger, that was derived from a lexicon named Qabas [55]. A POS tagger was enhanced using a smoothing lexicon model [121]. Moreover, a POS tagger algorithm was built using Brill's transformation-based tagger, trained on a lexicon of over 4,000,000 tokens manually annotated for Egyptian Arabic [122]. A large Arabic named entity lexicon

was automatically built using Arabic WordNet and Arabic Wikipedia [123]. A combination of lexicon-driven and statistical methods was applied for Arabic NER [124]. There is also a bilingual named entity lexicon, consisting of Arabic and English, that contains over 48,000 named entity pairs [125]. An enhanced Arabic NER technique using gold-standard and bootstrapped noisy features, including lexical features, was developed [126]. ArSenL is an Arabic sentiment lexicon derived from existing resources: ESWN, Arabic WordNet, and the Standard Arabic Morphological Analyser [127]. Arab-ESL is an Arabic emoji sentiment lexicon that interprets sentiments from emojis and compares them with their European counterparts to identify cultural variations [128]. An integrated sentiment lexicon with domain ontology was used as a feature approach for Arabic sentiment analysis [129]. Another study considered the critical role and enhancement of lexicons in dialectal Arabic sentiment analysis and expert validation [130]. Emo-SL is an emoji sentiment lexicon obtained from 58,000 Arabic tweets, and it is applied as a feature of a machine learning algorithm for sentiment analysis [131]. Online learning was analysed through Arabic tweets during the COVID-19 pandemic using the National Research Council Canada's word–emotion lexicon to reveal sentiments [132]. Many other studies have constructed or used a lexicon for Arabic sentiment analysis (e.g., [133,134]).

4.5. Sentiment Analysis

Sentiment analysis is one of the NLP tasks that have been widely explored in Arabic. It involves identifying the sentiment or emotion from a row of text. Sentiment analysis can be divided into document, sentence, phrase, and aspect levels. Document-level sentiment analysis identifies the sentiment or emotion from an entire document; most existing studies have applied this technique [135]. Sentence-level sentiment analysis involves determining the sentiment or emotion from a given sentence, and several studies have applied it [136]. Phrase-level sentiment analysis extracts the opinion or emotion from a collection of related words, and it has also been used in a number of studies [137]. Aspect-level sentiment analysis focuses on the overall feeling about a particular thing or aspect. It has been widely adopted in Arabic sentiment analysis research, with many studies employing it [138]. In addition, various studies have compared the different levels. A comparison between document- and sentence-level sentiment analysis can be found in [139]. A comparative analysis exploring the effectiveness of character-, sub-character-, and word-level Arabic sentiment analysis was conducted in [140]. A discriminative study of phrase- and word-level Arabic sentiment analysis was performed in [141]. A combined feature representation of both character- and word-level Arabic sentiment analysis was achieved in [142].

Sentiment analysis based on algorithm approaches and techniques can be grouped into lexicon-based, basic machine learning, deep learning, and hybrid models. Since the early research on Arabic sentiment analysis, most authors have been utilising the lexicons of dictionaries to measure sentences from text [139]. Subsequently, most researchers started using basic machine learning algorithms, such as SVM [143], naïve Bayes [144], k-nearest neighbours [145], decision tree [146], maximum entropy [147], and logistic regression [148], as well as other methods with different feature selection, including term frequency TF [149], term frequency-inverse document frequency TF-IDF [150], and lexicons [151]. Deep learning models are another type of machine learning algorithm that use neural networks with different layers [152]. They vectorise the textual data to be fed to the input layer, and the most commonly used technique is word embeddings [153], including Word2Vec [154], GloVe [155], and fastText [156]. Many research papers have applied deep learning models extensively, using different neural networks, such as deep belief network and deep auto encoder models [157], convolutional neural networks (CNNs) [158], recur-

rent neural networks (RNNs) [159], gated recurrent units (GRUs) [160], and long short-term memory (LSTM) [161]. Moreover, Arabic transformer-based models, which represent text data based on a contextual embedding BERT, are used, and AraBERT is one of the most well-known Arabic models [162]. Different transformer models have been applied along with different deep learning models for sentiment analysis purposes [163]. Transfer learning is the approach of reusing a model that was pretrained on one task to solve another related, similar task [164]. It has been applied in many studies [165]. Finally, there has been a notable growth in research on integrated models for Arabic sentiment analysis. These integrate different machine learning algorithms, such as CNN and LSTM [166]. Furthermore, CNN and Bi-LSTM for feature selection along with an SVM classifier has been proposed for Arabic sentiment analysis [167]. A hybrid semantic orientation lexical-based classifier and SVM algorithm has been introduced as well [168].

The number of publicly available Arabic sentiment datasets has been increasing, owing to the growing attention devoted to this field. The Arabic Sentiment Tweets Dataset (ASTD) contains approximately 54,000 tweets, and it is categorised into four classes: objective, subjective positive, subjective negative, and subjective mixed [169]. SemEval-2016 introduced various datasets covering multiple languages, including an Arabic dataset for hotel reviews consisting of over 1300 tweets classified as positive, negative, or neutral [170]. SemEval-2017 introduced a dataset of over 10,000 tweets in various Arabic dialects, including Levantine, Gulf, Egyptian, and Moroccan, with classification levels of two, three, and five classes [171]. A Twitter dataset for Arabic sentiment analysis was built with 2000 tweets that were equally split into positive and negative classes [172]. The Large-Scale Arabic Book Review (LABR) dataset has approximately 63,000 book reviews with five scale classes and is used for sentiment analysis purposes [173]. The Hotel Arabic Reviews Dataset (HARD) comprises over 400,000 hotel reviews that were collected from booking.com, and it has five labels [174]. The Arabic Health Services (Main-AHS and Sub-AHS) dataset has positive and negative classes; it was retrieved from Twitter, with Main-AHS comprising 2026 tweets [148] and Sub-AHS comprising 1732 tweets [175]. An Arabic benchmark dataset for sentiment analysis was created, comprising over 151,000 tweets in various Arabic dialects, grouped into two balanced classes: positive and negative [176]. The AraSenTi-Tweet dataset includes over 17,000 entities, which were manually classified into five categories: positive, negative, mixed, neutral, and indeterminate [177].

Extensive research in Arabic sentiment analysis has investigated various Arabic dialects and different applications. MSA has gained significant attention from many researchers compared to other Arabic dialects [178]. The sentiment aspects of the Sudanese dialect were examined by introducing two benchmark datasets, namely SudSenti2 (two classes) and SudSenti3 (three classes), and employing a CNN-based model [179]. A study performed sentiment analysis for the Iraqi dialect using Doc2Vec, trained on a large Iraqi Arabic corpus, and found that logistic regression and SVM outperformed other classifiers [180]. A framework was proposed for Moroccan Arabic tweet sentiment analysis, incorporating preprocessing techniques [181]. The sentiment analysis of the Bahraini dialect involved developing a balanced dataset of Bahraini dialects and applying transfer learning [182]. Furthermore, for sentiment analysis of the Algerian dialect, Word2Vec and TF-IDF were applied with SVM and LSTM models [183]. A comparison sentiment analysis study of the Saudi dialect between LSTM and Bi-LSTM models and SVM was conducted [183]. The sentiment of the Egyptian dialect was analysed at the sentence level using a combination of machine learning and semantic orientation features alongside a simple negation detection method [184]. Another study focused on the Emirati dialect based on a novel dataset of over 70,000 comments and applied TF-IDF, multiple machine learning classifiers, and an ensemble model [185]. A sentiment analysis model was pro-

posed for Jordanian Arabic dialect tweets, implementing SVM and naïve Bayes classifiers with TF-IDF-based features [186]. The sentiment analysis of the Lebanese dialect utilised transfer learning with XLM-RoBERTa—a multilingual pretrained model fine-tuned on English [187]. The sentiment of the Palestinian dialect was evaluated using a lexicon-based approach [188]. For the Tunisian dialect, sentiment analysis was conducted using machine learning to classify the polarity of comments [189]. Moreover, Arabic sentiment analysis targets special aspects or applications, such as politics [190], finance [191], customer reviews [192], health [193], and education and e-learning [132]. Some events, such as agricultural festivals in Al-Baha, Saudi Arabia [194], road traffic congestion [195], tourism and leisure [196], and the 2022 FIFA World Cup [197], were also analysed.

4.6. Text Classification

Arabic text classification resembles sentiment analysis in terms of levels, algorithms, and methodology. However, text classification typically differs from sentiment analysis in a wide range of datasets, in that it is inclusive of multiple label classes as well as their applications. Kaleej-2004 [198] has 5690 documents with four classes: economy, international news, local news, and sports. The Arabic Newspaper Archives dataset consists of 1445 documents grouped into nine categories: computer, economics, education, engineering, law, medicine, politics, religion, and sports [199]. The OSAC dataset is organised into ten categories, namely economy, history, education and family, religion and fatwa, sports, health, astronomy, law, stories, and food recipes, and comprises 22,428 documents [200]. The Watan-2004 dataset has 20,291 documents and is divided into six categories: culture, economy, international, local, religion, and sports [201]. The TALAA dataset has 57,827 instances distributed across eight classes: culture, economics, politics, religion, society, sports, world, and other [202]. The ANT dataset is divided into nine thematic categories, namely culture, diverse, economy, international news, local news, politics, society, sports, and technology, and contains 10,161 documents [203]. The Online Newspapers dataset has five news categories, namely sport, politics, culture, economy, and diverse, and the total number of documents is 111,728 [204]. The Arabic News dataset contains 6000 documents organised into four main categories: art, economy, accident, and politics [205]. The SANAD dataset provides a collection of over 190,000 documents, classified into seven groups: culture, finance, medical, politics, religion, sports, and technology [206]. RTAnews is a corpus of 23,837 Arabic news articles distributed among 40 labels [207]. NADiA consists of 451,230 articles covering 24 classes [208]. The News Portals multilabel dataset consists of over 29,000 articles categorised into four labels: Middle East, business, technology, and sports [209]. An Arabic tweets multi-label collection of 160,870 tweets covers four categories: sports, accidents, arts, and business [210]. With more than 500,000 articles, the Arabic news article dataset (ANAD) covers various topics, such as sports, local news, politics, economics, technology, tourism, entertainment, cars, health, and art [211]. WiHArD comprises 6027 hierarchical Arabic articles from Wikipedia, covering 12 categories across culture, history, math, and related subfields [212]. The Arabic sarcasm detection dataset (ArSarcasm) is a reannotated Arabic dataset of 10,547 tweets, of which 16% are sarcastic, with additional sentiment and dialect annotations [213]. The Arabic Functional Text Dimensions (AFTD) Corpus is a publicly available dataset of 3400 documents in 17 categories for classification tasks. Arabic news article classification dataset ANACD—a subset of ANAD—is a balanced Arabic news article classification dataset that consists of nine categories, namely tourism, economy, cars, technology, art, health, sports, local news, and politics, in which each class has 10,000 documents [214,215].

4.7. Text Summarisation

Arabic text summarisation is the process of automatically condensing Arabic texts while preserving their main ideas, semantic meaning, and linguistic integrity [216]. There are three main approaches to text summarisation: extractive, abstractive, and hybrid [217]. Extractive text summarisation techniques focus on selecting the most important sentences or phrases from the original text, using a number of methods [218]. First, statistical-based methods apply the statistical analysis approach, including frequency of occurrence, positional attributes, sentence length, and similarity of title, to sentence metrics to identify the most significant sentences and words. This approach has been combined with a semantic model using Word2Vec for Arabic text summarisation [219]. Second, graph-based methods conceptualise sentences as nodes, with edges denoting similarities between them within a weighted graph. They apply algorithms such as TextRank and LexRank to rank and extract the most important and discriminative sentences [220]. Finally, concept-based or semantic-based approaches consider capturing the meaning of the text by applying various techniques, such as word embeddings and semantic similarity, and they can be integrated with other approaches [219]. Abstractive text summarisation techniques involve comprehending the content and rephrasing a condensed version of the text instead of selecting existing sentences [221]. Early approaches included graph-based and semantic modelling [221,222], as well as rule-based modelling [223]. Subsequently, machine-learning-based approaches for text summarisation were applied to summarise Arabic text using basic machine learning algorithms [224], deep learning algorithms (e.g., CNN [225], BiLSTM, and AraBERT [226]), and RNN-based and BERT2BERT-based encoder–decoder models [226]. Finally, hybrid approaches were developed; they employ a combination of two or more methods to leverage their respective advantages for better results [227].

There are many Arabic text summarisation datasets. The Essex Arabic Summaries Corpus (EASC) has 150 articles, along with 765 summaries that were manually created [228]. The Large-Scale Arabic News Summarisation (LANS) Corpus was collected from 22 Arabic newspapers between 1999 and 2019, with over 8 million new articles and 1000 manually created summaries [229]. SumArabic contains over 84,000 [230] cross-lingual Arabic summarisation datasets, consisting of 21,000 articles [231]. The Large-Scale Multilingual Abstractive Summarisation for 44 Languages (the XL-Sum) was created from 44 languages and has 46,897 Arabic articles with human-generated summaries. The Arabic Headline Summary (AHS) has 300,000 articles, and their titles are considered their abstractive summaries [232]. Furthermore, there is a range of evaluation techniques for text summarisation that have been employed to assess Arabic text summarisation. They can be categorised into extrinsic metrics and intrinsic automatic metrics [233]. In intrinsic methods, the evaluation process compares the machine's summarisation with a human-generated summary of the text [233]. For instance, the Recall-Oriented Understudy for Gisting Evaluation (ROUGE) is a widely used set of metrics for evaluating automatic text summarisation techniques [234]. It has various methods: ROUGE-N, ROUGE-L [235], ROUGE-S, ROUGE-SU [236], and ROUGE-W [237]. Other examples are precision, which measures the content relevance of the generated summary; recall, which captures the coverage of the reference; and F-measure, which provides harmonic balance [238]. The Bilingual Evaluation Understudy (BLEU) [239] is another method. In contrast, in extrinsic methods, the evaluation of the summary is tested for downstream tasks, such as question answering or information retrieval [233].

4.8. Question Answering

Question answering in Arabic is an essential subfield of NLP, and it is an advanced version of information retrieval, in that it provides a precise answer rather than a list of doc-

uments [240]. There are various types of Arabic question answering: factoid, definitional, list, confirmation, complex, semantic, and open-domain question answering systems. The factoid type provides answers that are brief factual information, such as names, dates, and locations [241]. Definitional question answering involves seeking definitions or explanations, and it delivers brief descriptive answers [242]. List question answering is a rare type that views the input question as a 'bag of words' and then lists multiple possible answers or documents [243]. Confirmation question answering focuses on confirmation or denial responses—that is, 'yes or no' responses [244]. Causal or complex question answering systems justify causations by providing descriptive solutions to answer 'why' queries [245]. Machine reading comprehension (MRC) and open-domain question answering entail extracting or retrieving the answers from passages or unstructured text [246].

One of the challenges in this field is the scarcity of high-quality datasets. However, there are a number of available datasets for Arabic question answering. A dataset for Arabic why question answering system (DAWQAS) is an Arabic question answering dataset for 'why' questions, consisting of 3205 items generated from web documents [247]. HajjFQA is a dataset that was proposed for question answering in order to develop HajjBot and thus provide answers related to Hajj fatwas [248]. Arabic reading comprehension dataset (ARCD) consists of 1395 questions; it was crowdsourced from Wikipedia [246]. In addition, Arabic-SQuAD is an Arabic dataset that was machine-translated from the stanford question answering dataset (SQuAD) [246]. Translated cross language evaluation forum CLEF and text retrieval conference (TREC) are translated question answering datasets based on scientific papers and web pages, and it contains 2264 questions [249]. Finally, the Arabic Question–Answer Dataset (AQAD) has over 17,000 questions and answers retrieved from Arabic Wikipedia articles [250].

There are various techniques to assess the performance of question answering methods. Exact match is a comparison percentage that measures the exact match. It is the gold standard and can be used in factoid as well as MRC and open-domain question answering [251]. Accuracy measurement is used in factoid and yes/no questions [252]. Precision [251], recall [251], and F1-measure [251] are common evaluation measurements in factoid and list question answering. Furthermore, ROUGE, BLEU, and human evaluation have been applied for MRC and open-domain question answering [248].

4.9. Machine Translation

Machine translation of Arabic is another subfield of NLP; it aims to automate the process of translating Arabic text into other languages, or vice versa [253], as well as the translation of different Arabic dialects into each other [254]. This field shares the challenges mentioned in Section 3. An additional challenge impacting machine translation is code-switching, which refers to switching between Arabic words and other words from different languages within the text, posing issues for monolingual translation models [255]. The initial approaches in this field began with rule-based machine translation, which requires a set of linguistic rules for both languages, such as syntactic rules [256], morphological analysers [257], or bilingual dictionaries [258]. Then statistical machine translation methods emerged; these involve constructing a coherent statistical model based on sentence-aligned or phrase-aligned sets for both languages [259]. The next technique developed was neural machine translation, which is based on deep neural network algorithms, typically employing an encoder–decoder architecture with attention mechanisms to handle sequential text dependencies [260]. The most recent machine translation techniques are pretrained models and LLMs, which have altered the direction of this field. These integrated models employ cross-lingual transfer and extensive large-scale multilingual translation, such as Arabic–English translation [261] or the translation of Arabic dialects into MSA [262]. In addition,

a hybrid approach that is a combination of the different mentioned techniques has been applied [263].

A considerable collection of datasets relevant to Arabic machine translation is available. The Arabic language was included, along with five other official United Nations languages, in the United Nations Parallel Corpus v1.0 [264], which provides sentence pairs per language pair. The Qatar Arabic Language Bank (QALB) is another Arabic–English-aligned corpus [265]. The multi Arabic dialect applications and resources (MADAR), an Arabic dialect corpus, is a benchmark dataset that covers 25 Arabic dialects and MSA, with alignment of English [266]. Furthermore, the online social network–based multidialectal Arabic dataset (OSN-MDAD) is an English-to-multidialectal-Arabic translation dataset that provides contextual translation resources [267]. A multi-dialect Arabic to MSA Dial2MSA is another Arabic machine translation dataset for four Arabic dialects (Gulf, Egyptian, Levantine, and Maghrebi), evaluated using Seq2Seq models [268].

Various evaluation metrics have been used for Arabic–English translation. Error analysis measures the weakness of the machine translation [269]. A form of this is linguistic error analysis, which focuses on the linguistic structure [270]. Another assessment method is BLEU, which measures the similarities between the machine translation output and the reference translation [253]. AL-BLEU is a version of BLEU modified to consider Arabic's rich morphology [271]. Translation edit rate (TER) measures the minimum changes required of the output to match the reference translation [272]. METEOR evaluates the semantic and linguistic features between the machine output and the target translations using measurement scores, precision, and recall, with a penalty for fragmented matches [273]. Other automated evaluation methods are chrF, which checks character n-gram overlap, and chrF++, which measures similarity based on character- and word-level n-grams [274].

4.10. Large Language Models (LLMs)

LLMs, which have garnered growing interest in the past five years, represent a breakthrough in artificial intelligence (AI) technology in general and NLP in particular. LLMs are trained on immense amounts of text data in order to understand and generate human language in a sophisticated textual form [261]. They are primarily transformer-based models and developed to glean the structures, patterns, and nuances of human language from vast resources. There are three well-known architectural deep learning models for LLMs. First, BERT [275] has also been adapted for Arabic through models. For example, AraBERT is one of the earliest models [162]. ARBERT is focused on MSA, MARBERT was trained on a Twitter dataset [276], and CAMeLBERT was developed by CAMeL Lab and supports MSA and dialects [277]. Second, generative pretrained transformer (GPT) [278] and some Arabic models have been developed: AraGPT2 and AraGPT2-mega were trained on Arabic news, web, and social media [279], and ArabianGPT increases Arabic morphology and syntax capturing by reducing English tokens [280]. Finally, text-to-text and generation (T5) [281] has inspired the development of Arabic models: AraT5 was trained on large amounts of MSA and Twitter data [282], and AraMUS is the first Arabic pretrained language model with multibillion parameters [283].

Moreover, LLMs can be classified as monolingual, bilingual, and multilingual for Arabic. Monolingual models are those trained exclusively on Arabic textual data to capture morphology, syntax, and dialectal details from the trained text [284]. Examples include AraBERT [162], MARBERT [276], AraGPT2 [279], JASMINE [285], and CAMeLBERT [277]. Other specific dialectical monolingual models are AraRoBERTa for the Saudi, Egyptian, Kuwaiti, Omani, Lebanese, Jordanian, and Algerian dialects [286]; SaudiBERT for the Saudi dialect [287]; SudaBERT for the Sudanese dialect [288]; DziriBERT for the Algerian dialect [289]; MorRoBERTa, MorrBERT [290], DarijaBERT [291], and Atlas-Chat [292]

for the Moroccan dialect; TunBERT for the Tunisian dialect [293]; EgyBERT for the Egyptian dialect [294]. In addition, AlcLaM covers different Arabic dialects and is applied for offensive language detection and dialect identification [295]. Bilingual models apply dual corpora containing different languages, with the most common being Arabic and English [296]. For instance, GigaBERT is designed for information extraction in Arabic and English [297]. Moreover, ALLaM considers language alignment in training the model [261]. Others include Jais-chat, trained using 13 billion parameters [298], and AceGPT, trained on distinct Arabic cultural contexts (localisation issues) [299]. In multilingual models, large-scale multilingual corpora, including Arabic, are used to train the models, thereby providing cross-lingual generalisation. For example, AraLLaMA can align low-resource languages using large-scale models [300]. In addition, AraT5 utilises a sequence-to-sequence technique [282], and Fanar has Fanar Star and Fanar Prime for an Arabic multimodal LLM [301].

These LLMs show remarkable achievements in terms of understanding and generating text. Moreover, they have extended their capabilities to deal with other tasks, such as answering questions [242], summarising information [302], and translating languages and dialects [262]. These tasks have been used in the evaluation of Arabic LLMs [162]. However, researchers have proposed different evaluation benchmarks for LLMs due to the varying aspects of employing the models. These benchmarks include the fifth Nuanced Arabic Dialect Identification Shared Task (NADI 2024) [303] as well as AL-QASIDA [304]. In addition, a platform for benchmarking Arabic LLMs (BALSAM), covering 78 tasks across 14 categories, was developed [305]. Moreover, Safeguard can evaluate Arabic-region-specific safety, covering various sensitive topics comprising cultural, political, and social issues [306]. Massive multitask language understanding (MMLU) functions similarly, with a focus on cultural context [307]. AraTrust is an evaluation trustworthiness benchmark containing 522 human-written multiple-choice questions [308].

5. Discussion

This study reveals that NLP has advanced considerably with the rise of deep learning and pretrained language models. However, fundamental challenges remain due to the Arabic language's complex morphology, diglossia, and dialectal diversity. Most existing research focuses on MSA, leaving dialectal varieties underrepresented and limiting model generalisation across real-world contexts. Data scarcity and the lack of large, high-quality, and consistently annotated resources continue to hinder progress, especially in low-resource dialects. Furthermore, the absence of standardised evaluation benchmarks complicates fair comparison between models and tasks. Despite these challenges, recent trends in transfer learning, cross-lingual modelling, and the development of open-source Arabic corpora have shown promise in improving performance and accessibility. To ensure sustainable progress, future Arabic NLP research must prioritise dialectal inclusion, dataset standardisation, ethical fairness, and the creation of comprehensive benchmarks that reflect the linguistic and cultural diversity of the Arabic language.

6. Conclusions and Future Works

6.1. Conclusions

Arabic NLP stands at a critical intersection between linguistic complexity and technological innovation. This paper reveals that, despite notable progress enabled by deep learning and pretrained models, Arabic's morphological richness and dialectal diversity, in addition to its intricate orthography and ambiguity, continue to pose significant challenges for computational analysis. The effectiveness of Arabic NLP tasks often depends on limited datasets and approaches. However, integrating multiple techniques has improved

the performance. Furthermore, the emergence of LLMs has introduced new directions for all other tasks and has proven to be outstanding in understanding the Arabic language.

6.2. Future Works

In the context of Arabic NLP, future research consideration should tend to an urgent need for dialectally inclusive resources, standardised benchmarks, and evaluation frameworks remains. It is clear that there are challenges and gaps in working with multiple dialects that lack standardisation. Therefore, future research should focus on bridging the various dialects with MSA rather than considering a single dialect, to enhance machine understanding of Arabic. The effectiveness of LLMs in addressing the complex linguistic and morphological challenges of Arabic is evident; however, there remains a need for high-quality datasets across various domains, such as politics, economics, sports, and others. Moreover, collaborative initiatives between academia and industry can accelerate resource creation and promote fairness across different Arabic communities. Ultimately, bridging the gap between linguistic depth and computational efficiency will promote the evolution of Arabic NLP in terms of inclusiveness, domain impact capability, diverse cultural and dialectal coverage, and technological advancement across the Arabic language.

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References

1. Boudad, N.; Faizi, R.; Oulad Haj Thami, R.; Chiheb, R. Sentiment analysis in Arabic: A review of the literature. *Ain Shams Eng. J.* **2018**, *9*, 2479–2490. [CrossRef]
2. Moher, D.; Shamseer, L.; Clarke, M.; Ghersi, D.; Liberati, A.; Petticrew, M.; Shekelle, P.; Stewart, L.A. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst. Rev.* **2015**, *4*, 1. [CrossRef] [PubMed]
3. Khaliq, B.; Carroll, J. Unsupervised Induction of Arabic Root and Pattern Lexicons using Machine Learning. In Proceedings of the International Conference Recent Advances in Natural Language Processing RANLP 2013, Hissar, Bulgaria, 9–11 September 2013; pp. 350–356.
4. Al-Huri, I. Arabic Language: Historic and Sociolinguistic Characteristics. *Engl. Lit. Lang. Rev.* **2015**, *1*, 28–36. [CrossRef]
5. Mbarki, S.; Mourchid, M.; Silberztein, M. (Eds.) Formalizing Natural Languages with NooJ and Its Natural Language Processing Applications. In Proceedings of the 11th International Conference, NooJ 2017, Kenitra and Rabat, Morocco, 18–20 May 2017; Revised Selected Papers; Communications in Computer and Information Science, 1st ed.. Springer International Publishing: Cham, Switzerland, 2018; ISBN 978-3-319-73420-0.
6. Ryding, K.C. *A Reference Grammar of Modern Standard Arabic*, 1st ed.; Cambridge University Press: Cambridge, UK, 2005; ISBN 978-0-521-77771-1.
7. McCarthy, J.J. A Prosodic Theory of Nonconcatenative Morphology. *Linguist. Inq.* **1981**, *12*, 373–418.
8. Al-Kabi, M.N.; Kazakzeh, S.A.; Abu Ata, B.M.; Al-Rababah, S.A.; Alsmadi, I.M. A novel root based Arabic stemmer. *J. King Saud Univ. Comput. Inf. Sci.* **2015**, *27*, 94–103. [CrossRef]
9. Al-Sughaiyer, I.A.; Al-Kharashi, I.A. Arabic morphological analysis techniques: A comprehensive survey. *J. Am. Soc. Inf. Sci.* **2004**, *55*, 189–213. [CrossRef]
10. Owens, J. (Ed.) *The Oxford Handbook of Arabic Linguistics*, 1st ed.; Oxford University Press: Oxford, UK, 2013; ISBN 978-0-19-976413-6.
11. Diab, M.; Ghoneim, M.; Habash, N. Arabic diacritization in the context of statistical machine translation. In Proceedings of the Machine Translation Summit XI: Papers, Copenhagen, Denmark, 10 September 2007.
12. Saiegh-Haddad, E.; Henkin-Roitfarb, R. The Structure of Arabic Language and Orthography. In *Handbook of Arabic Literacy*; Saiegh-Haddad, E., Joshi, R.M., Eds.; Literacy Studies; Springer: Dordrecht, The Netherlands, 2014; Volume 9, pp. 3–28, ISBN 978-94-017-8544-0.
13. Maroun, M.; Hanley, J.R. Diacritics improve comprehension of the Arabic script by providing access to the meanings of heterophonic homographs. *Read. Writ.* **2017**, *30*, 319–335. [CrossRef]

14. Aabed, M.A.; Awaideh, S.M.; Elshafei, A.-R.M.; Gutub, A.A. Arabic Diacritics based Steganography. In Proceedings of the 2007 IEEE International Conference on Signal Processing and Communications, Dubai, United Arab Emirates, 24–27 November 2007; pp. 756–759.
15. Midhwah, A.A.; Alhawary, M.T. Arabic Diacritics and Their Role in Facilitating Reading Speed, Accuracy, and Comprehension by English L2 Learners of Arabic. *Mod. Lang. J.* **2020**, *104*, 418–438. [CrossRef]
16. Chennoufi, A.; Mazroui, A. Morphological, syntactic and diacritics rules for automatic diacritization of Arabic sentences. *J. King Saud Univ. Comput. Inf. Sci.* **2017**, *29*, 156–163. [CrossRef]
17. Boumaraf, A.; Bekal, S.; Macoir, J. The Orthographic Ambiguity of the Arabic Graphic System: Evidence from a Case of Central Agraphia Affecting the Two Routes of Spelling. *Behav. Neurol.* **2022**, *2022*, 8078607. [CrossRef]
18. Hegazi, M.O.; Al-Dossari, Y.; Al-Yahy, A.; Al-Sumari, A.; Hilal, A. Preprocessing Arabic text on social media. *Heliyon* **2021**, *7*, e06191. [CrossRef] [PubMed]
19. Farghaly, A.; Shaalan, K. Arabic Natural Language Processing: Challenges and Solutions. *ACM Trans. Asian Lang. Inf. Process.* **2009**, *8*, 14. [CrossRef]
20. Abu Elhija, D. A new writing system? Developing orthographies for writing Arabic dialects in electronic media. *Writ. Syst. Res.* **2014**, *6*, 190–214. [CrossRef]
21. Saadane, H.; Habash, N. A Conventional Orthography for Algerian Arabic. In Proceedings of the Second Workshop on Arabic Natural Language Processing, Beijing, China, 26–31 July 2015; pp. 69–79.
22. Cinta, F.; Irawan, B.; Hasan, N. Semantic study: Polysemi in Arabic form of verb and noun. *AKSARA J. Bhs. Dan Sastra* **2023**, *24*, 520–530. [CrossRef]
23. Mohammed Abdul-Ghafour, A.-Q.K.; Mat Awal, N.; Zainudin, I.S.; Aladdin, A. The Interplay of Qur'ānic Synonymy and Polysemy with Special Reference to Al-asfār and Al-kutub (the Books) and their English Translations. *3L Southeast Asian J. Engl. Lang. Stud.* **2019**, *25*, 129–143. [CrossRef]
24. Al-Lahham, Y.A. Index Term Selection Heuristics for Arabic Text Retrieval. *Arab. J. Sci. Eng.* **2021**, *46*, 3345–3355. [CrossRef]
25. Al-shameri, N.; Al-Khalifa, H. Arabic paraphrased parallel synthetic dataset. *Data Brief* **2024**, *57*, 111004. [CrossRef]
26. Boujou, E.; Chataoui, H.; Mekki, A.E.; Benjelloun, S.; Chairi, I.; Berrada, I. An open access NLP dataset for Arabic dialects: Data collection, labeling, and model construction. *arXiv* **2021**, arXiv:2102.11000. [CrossRef]
27. Fadel, A.; Tuffaha, I.; Al-Ayyoub, M. Neural Arabic Text Diacritization: State-of-the-Art Results and a Novel Approach for Arabic NLP Downstream Tasks. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* **2022**, *21*, 1–25. [CrossRef]
28. Al-Ayyoub, M.; Nuseir, A.; Alsmearat, K.; Jararweh, Y.; Gupta, B. Deep learning for Arabic NLP: A survey. *J. Comput. Sci.* **2018**, *26*, 522–531. [CrossRef]
29. Aladeemy, A.A.; Alzahrani, A.; Algarni, M.H.; Alsubari, S.N.; Aldhyani, T.H.H.; Deshmukh, S.N.; Khalaf, O.I.; Wong, W.-K.; Aqburi, S. Advancements and challenges in Arabic sentiment analysis: A decade of methodologies, applications, and resource development. *Heliyon* **2024**, *10*, e39786. [CrossRef]
30. Bounhas, I. On the Usage of a Classical Arabic Corpus as a Language Resource: Related Research and Key Challenges. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* **2019**, *18*, 23. [CrossRef]
31. Weiss, B. A Theory of the Parts of Speech in Arabic (Noun, Verb and Particle): A Study in "ʿilm al-waḍʿ". *Arabica* **1976**, *23*, 23–36. [CrossRef]
32. Ditters, W.E. A Formal Approach to Arabic Syntax: The Noun Phrase and the Verb Phrase. Ph.D. Thesis, Nijmegen University, Nijmegen Luxor, The Netherlands, 1992.
33. McOmber, M.L. Morpheme edges and Arabic infixation. In *Current Issues in Linguistic Theory*; Eid, M., Ed.; John Benjamins Publishing Company: Amsterdam, The Netherlands, 1995; Volume 130, p. 173, ISBN 978-90-272-3633-3.
34. Attia, M.A. Arabic tokenization system. In Proceedings of the 2007 Workshop on Computational Approaches to Semitic Languages Common Issues and Resources-Semitic '07, Prague, Czech Republic, 28 June 2007; pp. 65–72.
35. Qaroush, A.; Abu Farha, I.; Ghanem, W.; Washaha, M.; Maali, E. An efficient single document Arabic text summarization using a combination of statistical and semantic features. *J. King Saud Univ. Comput. Inf. Sci.* **2021**, *33*, 677–692. [CrossRef]
36. Chennoufi, A.; Mazroui, A. Impact of morphological analysis and a large training corpus on the performances of Arabic diacritization. *Int. J. Speech Technol.* **2016**, *19*, 269–280. [CrossRef]
37. Azmi, A.M.; Alnefaie, R.M.; Aboalsamh, H.A. Light Diacritic Restoration to Disambiguate Homographs in Modern Arabic Texts. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* **2022**, *21*, 60. [CrossRef]
38. Vicente, Á. From stigmatization to predilection: Folk metalinguistic discourse on social media on the northwestern Moroccan Arabic variety. *Int. J. Sociol. Lang.* **2022**, *2022*, 133–154. [CrossRef]
39. Chennafi, M.E.; Bedlaoui, H.; Dahou, A.; Al-Qaness, M.A.A. Arabic Aspect-Based Sentiment Classification Using Seq2Seq Dialect Normalization and Transformers. *Knowledge* **2022**, *2*, 388–401. [CrossRef]
40. Abidin, Z.; Junaidi, A. Wamiliana Text Stemming and Lemmatization of Regional Languages in Indonesia: A Systematic Literature Review. *J. Inf. Syst. Eng. Bus. Intell.* **2024**, *10*, 217–231. [CrossRef]

41. Nazir, S.; Asif, M.; Rehman, M.; Ahmad, S. Machine learning based framework for fine-grained word segmentation and enhanced text normalization for low resourced language. *PeerJ Comput. Sci.* **2024**, *10*, e1704. [CrossRef] [PubMed]
42. Khoja, S.; Garside, R. *Stemming Arabic Text*; Computing Department, Lancaster University: Lancaster, UK, 1999; p. 29.
43. Al-Kabi, M.N. Towards improving Khoja rule-based Arabic stemmer. In Proceedings of the 2013 IEEE Jordan Conference on Applied Electrical Engineering and Computing Technologies (AEECT), Amman, Jordan, 3–5 December 2013; pp. 1–6.
44. Larkey, L.S.; Ballesteros, L.; Connell, M.E. Light Stemming for Arabic Information Retrieval. In *Arabic Computational Morphology*; Soudi, A., Bosch, A.V.D., Neumann, G., Eds.; Text, Speech and Language Technology; Springer: Dordrecht, The Netherlands, 2007; Volume 38, pp. 221–243, ISBN 978-1-4020-6045-8.
45. Taghva, K.; Elkhoury, R.; Coombs, J. Arabic stemming without a root dictionary. In Proceedings of the International Conference on Information Technology: Coding and Computing (ITCC'05)-Volume II, Las Vegas, NV, USA, 4–6 April 2005; Volume 1, pp. 152–157.
46. Ghwanmeh, S.; Kanaan, G.; Al-Shalabi, R.; Rabab'ah, S. Enhanced Algorithm for Extracting the Root of Arabic Words. In Proceedings of the 2009 Sixth International Conference on Computer Graphics, Imaging and Visualization, Tianjin, China, 11–14 August 2009; pp. 388–391.
47. Kanan, T.; Fox, E.A. Automated Arabic text classification with P-Stemmer, machine learning, and a tailored news article taxonomy. *J. Assoc. Inf. Sci. Tech.* **2016**, *67*, 2667–2683. [CrossRef]
48. Al-Khatib, R.M.; Zerrouki, T.; Abu Shquier, M.M.; Balla, A. Tashaphyne0.4: A new Arabic light stemmer based on rhyzome modeling approach. *Inf. Retr. J.* **2023**, *26*, 14. [CrossRef]
49. Habash, N.; Rambow, O.; Roth, R. MADA + TOKAN: A toolkit for Arabic tokenization, diacritization, morphological disambiguation, POS tagging, stemming and lemmatization. In Proceedings of the Second International Conference on Arabic Language Resources and Tools, Cairo, Egypt, 22–23 April 2009; Volume 41, p. 62.
50. Boudchiche, M.; Mazroui, A. A hybrid approach for Arabic lemmatization. *Int. J. Speech Technol.* **2019**, *22*, 563–573. [CrossRef]
51. Rabiner, L.R. A tutorial on hidden Markov models and selected applications in speech recognition. *Proc. IEEE* **1989**, *77*, 257–286. [CrossRef]
52. Boudlal, A.; Lakhouaja, A.; Mazroui, A.; Meziane, A.; Bebah, M.; Shoul, M. Alkhalil morpho sys1: A morphosyntactic analysis system for Arabic texts. In Proceedings of the International Arab Conference on Information Technology, New York, NY, USA, 14–16 December 2010; pp. 1–6.
53. Boudchiche, M.; Mazroui, A.; Ould Abdallahi Ould Bebah, M.; Lakhouaja, A.; Boudlal, A. AlKhalil Morpho Sys 2: A robust Arabic morpho-syntactic analyzer. *J. King Saud Univ. Comput. Inf. Sci.* **2017**, *29*, 141–146. [CrossRef]
54. Jarrar, M.; Akra, D.; Hammouda, T. Alma: Fast Lemmatizer and POS Tagger for Arabic. *Procedia Comput. Sci.* **2024**, *244*, 378–387. [CrossRef]
55. Jarrar, M.; Hammouda, T.H. Qabas: An Open-Source Arabic Lexicographic Database. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024, Torino, Italy, 20–25 May 2024; pp. 13363–13370.
56. Maamouri, M.; Bies, A.; Kulick, S.; Gaddeche, F.; Mekki, W.; Krouna, S.; Bouziri, B.; Zaghouani, W. *Arabic Treebank: Part 2 V 3.1*; Linguistic Data Consortium: Philadelphia, PA, USA, 2011. [CrossRef]
57. Jarrar, M.; Malaysha, S.; Hammouda, T.; Khalilia, M. SALMA: Arabic Sense-Annotated Corpus and WSD Benchmarks. In Proceedings of the ArabicNLP 2023, Singapore (hybrid conference), 7 December 2023; pp. 359–369.
58. Lee, Y.-S.; Papineni, K.; Roukos, S.; Emam, O.; Hassan, H. Language model based Arabic word segmentation. In Proceedings of the 41st Annual Meeting on Association for Computational Linguistics-ACL '03, Sapporo, Japan, 7–12 July 2003; Volume 1, pp. 399–406.
59. El Isbihani, A.; Khadivi, S.; Bender, O.; Ney, H. Morpho-syntactic Arabic Preprocessing for Arabic to English Statistical Machine Translation. In Proceedings of the Workshop on Statistical Machine Translation, New York, NY, USA, 8–9 June 2006; pp. 15–22.
60. Hammouda, N.G.; Haddar, K. Integration of a Segmentation Tool for Arabic Corpora in NooJ Platform to Build an Automatic Annotation Tool. In Proceedings of the Automatic Processing of Natural-Language Electronic Texts with NooJ, České Budějovice, Czech Republic, 9–11 June 2016; pp. 89–100.
61. Silberztein, M.; Váradi, T.; Tadić, M. Open source multi-platform NooJ for NLP. In Proceedings of the COLING 2012, Mumbai, India, 8–15 December 2012; pp. 401–408.
62. Souri, A.; Al Achhab, M.; El Mouhajir, B.E. A Proposed Approach for Arabic Language Segmentation. In Proceedings of the 2015 First International Conference on Arabic Computational Linguistics (ACLing), Cairo, Egypt, 17–20 April 2015; pp. 43–48.
63. Monroe, W.; Green, S.; Manning, C.D. Word Segmentation of Informal Arabic with Domain Adaptation. In Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), Baltimore, MD, USA, 22–27 June 2014; pp. 206–211.
64. Eldesouki, M.; Samih, Y.; Abdelali, A.; Attia, M.; Mubarak, H.; Darwish, K.; Laura, K. Arabic Multi-Dialect Segmentation: Bi-LSTM-CRF vs. SVM. *arXiv* **2017**, arXiv:1708.05891. [CrossRef]

65. Cheragui, M.A.; Hiri, E. Arabic Text Segmentation using Contextual Exploration and Morphological Analysis. In Proceedings of the 2020 2nd International Conference on Mathematics and Information Technology (ICMIT), Adrar, Algeria, 18–19 February 2020; pp. 220–225.
66. Pasha, A.; Al-Badrashiny, M.; Diab, M.; El Kholy, A.; Eskander, R.; Habash, N.; Pooleery, M.; Rambow, O.; Roth, R. MADAMIRA: A Fast, Comprehensive Tool for Morphological Analysis and Disambiguation of Arabic. In Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14), Reykjavik, Iceland, 26–31 May 2014; pp. 1094–1101.
67. Diab, M. Second generation AMIRA tools for Arabic processing: Fast and robust tokenization, POS tagging, and base phrase chunking. In Proceedings of the 2nd International Conference on Arabic Language Resources and Tools, Cairo, Egypt, 22–23 April 2009; Volume 110, p. 198.
68. Abdelali, A.; Darwish, K.; Durrani, N.; Mubarak, H. Farasa: A Fast and Furious Segmenter for Arabic. In Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations, San Diego, CA, USA, 12–17 June 2016; pp. 11–16.
69. Darwish, K.; Mubarak, H. Farasa: A New Fast and Accurate Arabic Word Segmenter. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), Portorož, Slovenia, 23–28 May 2016; pp. 1070–1074.
70. Obeid, O.; Zalmout, N.; Khalifa, S.; Taji, D.; Oudah, M.; Alhafni, B.; Inoue, G.; Eryani, F.; Erdmann, A.; Habash, N. CAMeL tools: An open source python toolkit for Arabic natural language processing. In Proceedings of the Twelfth Language Resources and Evaluation Conference, Marseille, France, 13–15 May 2020; pp. 7022–7032.
71. Manning, C.; Surdeanu, M.; Bauer, J.; Finkel, J.; Bethard, S.; McClosky, D. The Stanford CoreNLP Natural Language Processing Toolkit. In Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics: System Demonstrations, Baltimore, MD, USA, 22–27 June 2014; pp. 55–60.
72. Benajiba, Y.; Diab, M.; Rosso, P. Arabic Named Entity Recognition using Optimized Feature Sets. In Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing, Honolulu, HI, USA, 25–27 October 2008; pp. 284–293.
73. Darwish, K. Named Entity Recognition using Cross-lingual Resources: Arabic as an Example. In Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Sofia, Bulgaria, 4–9 August 2013; pp. 1558–1567.
74. Salah, R.E.; Binti Zakaria, L.Q. Building the Classical Arabic Named Entity Recognition Corpus (CANERCorpus). In Proceedings of the 2018 Fourth International Conference on Information Retrieval and Knowledge Management (CAMP), Kota Kinabalu, Malaysia, 26–28 March 2018; pp. 1–8.
75. Al-Qurishi, M.S.; Souissi, R. Arabic Named Entity Recognition Using Transformer-based-CRF Model. In Proceedings of the 4th International Conference on Natural Language and Speech Processing (ICNLSP 2021), Trento, Italy, 12–13 November 2021; pp. 262–271.
76. Oudah, M.; Shaalan, K. A Pipeline Arabic Named Entity Recognition using a Hybrid Approach. In Proceedings of the COLING 2012, Mumbai, India, 8–15 December 2012; pp. 2159–2176.
77. Ahmed, S.; van der Goot, R.; Rehman, M.; Kruse, C.; Özsoy, Ö.; Mehler, A.; Roig, G. Tafsir Dataset: A Novel Multi-Task Benchmark for Named Entity Recognition and Topic Modeling in Classical Arabic Literature. In Proceedings of the 29th International Conference on Computational Linguistics, Gyeongju, Republic of Korea, 12–17 October 2022; pp. 3753–3768.
78. Albahli, S. An Advanced Natural Language Processing Framework for Arabic Named Entity Recognition: A Novel Approach to Handling Morphological Richness and Nested Entities. *Appl. Sci.* **2025**, *15*, 3073. [CrossRef]
79. Dahou, A.; Abd Elaziz, M.; Mohamed, H.; Dahou, A.H.; Al-Qaness, M.A.A.; Ghetas, M.; Ewess, A.; Zheng, Z. Linguistic feature fusion for Arabic fake news detection and named entity recognition using reinforcement learning and swarm optimization. *Neurocomputing* **2024**, *598*, 128078. [CrossRef]
80. Hkiri, E.; Mallat, S.; Zrigui, M. Arabic-English Text Translation Leveraging Hybrid NER. In Proceedings of the 31st Pacific Asia Conference on Language, Information and Computation, Cebu City, Philippines, 16–18 November 2017; pp. 124–131.
81. Asbayou, O. Automatic Arabic Named Entity Extraction and Classification for Information Retrieval. In Proceedings of the International Journal on Natural Language Computing, Zurich, Switzerland, 21–22 November 2020; Volume 9, pp. 1–22.
82. Sabty, C.; Elmahdy, M.; Abdennadher, S. Arabic Named Entity Recognition Using Clustered Word Embedding. In Proceedings of the Computational Linguistics and Intelligent Text Processing, La Rochelle, France, 7–13 April 2019; pp. 41–49.
83. Abouenour, L.; Bouzoubaa, K.; Rosso, P. IDRAAQ: New Arabic Question Answering system based on Query Expansion and Passage Retrieval. In Proceedings of the CLEF 2012—QA4MRE Workshop, CEUR Workshop Proceedings. Rome, Italy, 17–20 September 2012; Volume 1178.
84. Essa, N.; El-Gayar, M.M.; El-Daydamony, E.M. Enhanced model for abstractive Arabic text summarization using natural language generation and named entity recognition. *Neural Comput. Appl.* **2025**, *37*, 7279–7301. [CrossRef]
85. Abdelrahman, S.; Elarnaoty, M.; Magdy, M.; Fahmy, A. Integrated Machine Learning Techniques for Arabic Named Entity Recognition. *Int. J. Sci. Innov. Eng.* **2010**, *7*, 27–36.

86. Shaalan, K.; Raza, H. Person Name Entity Recognition for Arabic. In Proceedings of the 2007 Workshop on Computational Approaches to Semitic Languages: Common Issues and Resources, Prague, Czech Republic, 28 June 2007; pp. 17–24.
87. Alkharashi, I. Person named entity generation and recognition for Arabic language. In Proceedings of the Second International Conference on Arabic Language Resources and Tools, Cairo, Egypt, 22–23 April 2009; pp. 205–208.
88. Shaalan, K.; Raza, H. NERA: Named Entity Recognition for Arabic. *J. Am. Soc. Inf. Sci.* **2009**, *60*, 1652–1663. [CrossRef]
89. Habash, N.Y. *Introduction to Arabic Natural Language Processing*; Synthesis Lectures on Human Language Technologies; Springer International Publishing: Cham, Switzerland, 2010; ISBN 978-3-031-01011-8.
90. Zaraket, F.A.; Jaber, A. MATAr: Morphology-based Tagger for Arabic. In Proceedings of the 2013 ACS International Conference on Computer Systems and Applications (AICCSA), Ifrane, Morocco, 27–30 May 2013; pp. 1–4.
91. Albared, M.; Omar, N.; Ab Aziz, M.J. Developing a Competitive HMM Arabic POS Tagger Using Small Training Corpora. In Proceedings of the Intelligent Information and Database Systems ACIIDS 2011, Daegu, Korea, 20–22 April 2011; pp. 288–296.
92. AlGahtani, S.; Black, W.; McNaught, J. *Arabic Part-Of-Speech Tagging Using Transformation-Based Learning*; Citeseer: Cairo, Egypt, 2009; pp. 66–70.
93. El Hadj, Y.; Al-Sughayeir, I.; Al-Ansari, A. *Arabic Part-of-Speech Tagging Using the Sentence Structure*; Citeseer: Cairo, Egypt, 2009; pp. 241–245.
94. Attia, M.; Rashwan, M.A.A.; Al-Badrashiny, M.A.S.A.A. Fassieh, a Semi-Automatic Visual Interactive Tool for Morphological, PoS-Tags, Phonetic, and Semantic Annotation of Arabic Text Corpora. *IEEE Trans. Audio Speech Lang. Process.* **2009**, *17*, 916–925. [CrossRef]
95. Hamdi, A.; Nasr, A.; Habash, N.; Gala, N. POS-tagging of Tunisian Dialect Using Standard Arabic Resources and Tools. In Proceedings of the Second Workshop on Arabic Natural Language Processing, Association for Computational Linguistics. Beijing, China, 26–31 July 2015; pp. 59–68.
96. Nahar, K.M.O.; Al Eroud, A.F.; Barahoush, M.; Al-Akhras, A.M. SAP: Standard Arabic Profiling Toolset for Textual Analysis. *Int. J. Mach. Learn.* **2019**, *9*, 222–229. [CrossRef]
97. Kallas, O.; Inoue, G.; Habash, N. EMAD: A Bridge Tagset for Unifying Arabic POS Annotations. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), Torino, Italy, 20–25 May 2024; pp. 5637–5643.
98. Himdi, H.T.; Assiri, F.Y. Tasaheel: An Arabic Automative Textual Analysis Tool—All in One. *IEEE Access* **2023**, *11*, 139979–139992. [CrossRef]
99. Tnaji, K.; Bouzoubaa, K.; Aouragh, S.L. A Light Arabic POS Tagger Using a Hybrid Approach. In *Digital Technologies and Applications*; Motahhir, S., Bossoufi, B., Eds.; Lecture Notes in Networks and Systems; Springer International Publishing: Cham, Switzerland, 2021; Volume 211, pp. 199–208, ISBN 978-3-030-73881-5.
100. Saidi, R.; Jarray, F.; Mansour, M. A BERT Based Approach for Arabic POS Tagging. In *Advances in Computational Intelligence*; Rojas, I., Joya, G., Català, A., Eds.; Lecture Notes in Computer Science; Springer International Publishing: Cham, Switzerland, 2021; Volume 12861, pp. 311–321, ISBN 978-3-030-85029-6.
101. Al Shamsi, F.; Guessoum, A. A hidden Markov model-based POS tagger for Arabic. In Proceedings of the 8th International Conference on the Statistical Analysis of Textual Data, Besançon, France, 19–21 April 2006; pp. 31–42.
102. Ababou, N.; Mazroui, A. A hybrid Arabic POS tagging for simple and compound morphosyntactic tags. *Int. J. Speech Technol.* **2016**, *19*, 289–302. [CrossRef]
103. AlKhwiter, W.; Al-Twairesh, N. Part-of-speech tagging for Arabic tweets using CRF and Bi-LSTM. *Comput. Speech Lang.* **2021**, *65*, 101138. [CrossRef]
104. Saber, Y.M.; Abdel-Galil, H.; El-Fatah Belal, M.A. Arabic ontology extraction model from unstructured text. *J. King Saud Univ. Comput. Inf. Sci.* **2022**, *34*, 6066–6076. [CrossRef]
105. Yousif, S.A.; Samawi, V.W.; Elkabani, I.; Zantout, R. The effect of combining different semantic relations on Arabic text classification. *World Comput. Sci. Inform. Technol. J.* **2015**, *5*, 12–118.
106. Yousif, S.A.; Samawi, V.W.; Elkabani, I.; Zantout, R. Enhancement of Arabic text classification using semantic relations with part of speech tagger. *W Trans. Adv. Electr. Comput. Eng.* **2015**, 195–201.
107. Himdi, H.T. Classification of Arabic Real and Fake News Based on Arabic Textual Analysis. Ph.D. Thesis, University of Strathclyde, Glasgow, UK, 2022.
108. Alias, N.; Rahman, N.A.; Alias, M.N.; Nor, Z.M.; Ahmad, N.A.; Ismail, N.K. Tagging Algorithm and POS Tags for Narrator's Name in Hadith Document. In Proceedings of the 2023 4th International Conference on Artificial Intelligence and Data Sciences (AiDAS), Ipoh, Malaysia, 6–7 September 2023; pp. 126–130.
109. Nerabie, A.M.; AlKhatib, M.; Mathew, S.S.; Barachi, M.E.; Oroumchian, F. The Impact of Arabic Part of Speech Tagging on Sentiment Analysis: A New Corpus and Deep Learning Approach. *Procedia Comput. Sci.* **2021**, *184*, 148–155. [CrossRef]
110. Darwish, K.; Mubarak, H.; Abdelali, A.; Eldesouki, M. Arabic POS Tagging: Don't Abandon Feature Engineering Just Yet. In Proceedings of the Third Arabic Natural Language Processing Workshop, Valencia, Spain, 3 April 2017; pp. 130–137.

111. Huang, C.; Calzolari, N.; Gangemi, A.; Lenci, A.; Oltramari, A.; Prevot, L. (Eds.) *Ontology and the Lexicon: A Natural Language Processing Perspective*, 1st ed.; Cambridge University Press: Cambridge, UK, 2010; ISBN 978-0-521-88659-8.
112. Pustejovsky, J.; Boguraev, B. Lexical knowledge representation and natural language processing. *Artif. Intell.* **1993**, *63*, 193–223. [CrossRef]
113. Kwaik, K.A.; Saad, M.; Chatzikyriakidis, S.; Dobnik, S. A Lexical Distance Study of Arabic Dialects. *Procedia Comput. Sci.* **2018**, *142*, 2–13. [CrossRef]
114. Sawalha, M.; Atwell, E.; Abushariah, M.A.M. SALMA: Standard Arabic Language Morphological Analysis. In Proceedings of the 2013 1st International Conference on Communications, Signal Processing, and their Applications (ICCSPA), Sharjah, United Arab Emirates, 12–14 February 2013; pp. 1–6.
115. Smrž, O. ElixirFM—Implementation of Functional Arabic Morphology. In Proceedings of the 2007 Workshop on Computational Approaches to Semitic Languages: Common Issues and Resources, Prague, Czech Republic, 28 June 2007; pp. 1–8.
116. Buckwalter, T. *Buckwalter Arabic Morphological Analyzer Version 1.0*; Linguistic Data Consortium: Philadelphia, PA, USA, 2002. [CrossRef]
117. Buckwalter, T. *Buckwalter Arabic Morphological Analyzer Version 2.0*; Linguistic Data Consortium: Philadelphia, PA, USA, 2004. [CrossRef]
118. Koskenniemi, K. *Two-Level Morphology: A General Computational Model for Word-Form Recognition and Production*; Department of General Linguistics, University of Helsinki: Helsinki, Finland, 1983; ISBN 951-45-3201-5.
119. Beesley, K.R. Arabic finite-state morphological analysis and generation. In Proceedings of the 16th conference on Computational Linguistics, Copenhagen, Denmark, 5–9 August 1996; Association for Computational Linguistics: Copenhagen, Denmark, 1996; Volume 1, p. 89.
120. Gridach, M.; Chenfour, N. Developing a new system for Arabic morphological analysis and generation. In Proceedings of the 2nd Workshop on South Southeast Asian Natural Language Processing (WSSANLP), Chiang Mai, Thailand, 8–9 November 2011; pp. 52–57.
121. Manour, S.; Sima'an, K.; Winter, Y. Smoothing a lexicon-based pos tagger for Arabic and Hebrew. In Proceedings of the 2007 Workshop on Computational Approaches to Semitic Languages: Common Issues and Resources, Prague, Czech Republic, 28 June 2007; pp. 97–103.
122. Al-Sabbagh, R.; Girju, R. A supervised POS tagger for written Arabic social networking corpora. In Proceedings of the KONVENS2012—The 11th Conference on Natural Language Processing, Vienna, Austria, 19–21 September 2012; pp. 39–52.
123. Attia, M.; Toral, A.; Tounsi, L.; Monachini, M.; van Genabith, J. *An Automatically Built Named Entity Lexicon for Arabic*; European Language Resources Association: Valletta, Malta, 2010.
124. Halpern, J. Lexicon-driven approach to the recognition of Arabic named entities. In Proceedings of the Second International Conference on Arabic Language Resources and Tools, Cairo, Egypt, 22–23 April 2009; pp. 193–198.
125. Hkiri, E.; Mallat, S.; Zrigui, M.; Mars, M. Constructing a Lexicon of Arabic-English Named Entity using SMT and Semantic Linked Data. *Int. Arab J. Inf. Technol. (IAJIT)* **2017**, 14.
126. Benajiba, Y.; Zitouni, I.; Diab, M.; Rosso, P. Arabic Named Entity Recognition: Using Features Extracted from Noisy Data. In Proceedings of the ACL 2010 Conference Short Papers, Uppsala, Sweden, 11–16 July 2010; pp. 281–285.
127. Badaro, G.; Baly, R.; Hajj, H.; Habash, N.; El-Hajj, W. A large scale Arabic sentiment lexicon for Arabic opinion mining. In Proceedings of the EMNLP 2014 Workshop on Arabic Natural Language Processing (ANLP), Doha, Qatar, 25 October 2014; pp. 165–173.
128. Hakami, S.A.A.; Hendley, R.; Smith, P. Arabic Emoji Sentiment Lexicon (Arab-ESL): A Comparison between Arabic and European Emoji Sentiment Lexicons. In Proceedings of the Sixth Arabic Natural Language Processing Workshop, Kyiv, Ukraine, 19 April 2021; pp. 60–71.
129. Khabour, S.M.; Al-Radaideh, Q.A.; Mustafa, D. A New Ontology-Based Method for Arabic Sentiment Analysis. *Big Data Cogn. Comput.* **2022**, *6*, 48. [CrossRef]
130. Sherif, S.M.; Alamoodi, A.H. Lexicon annotation in sentiment analysis for dialectal Arabic: Consensus Expert Standardized Criteria. *Appl. Data Sci. Anal.* **2024**, *2024*, 165–172. [CrossRef]
131. Alfreihat, M.; Almousa, O.S.; Tashtoush, Y.; AlSobeh, A.; Mansour, K.; Migdady, H. Emo-SL Framework: Emoji Sentiment Lexicon Using Text-Based Features and Machine Learning for Sentiment Analysis. *IEEE Access* **2024**, *12*, 81793–81812. [CrossRef]
132. Ali, M.M. Arabic sentiment analysis about online learning to mitigate COVID-19. *J. Intell. Syst.* **2021**, *30*, 524–540. [CrossRef]
133. Al-Moslmi, T.; Albared, M.; Al-Shabi, A.; Omar, N.; Abdullah, S. Arabic senti-lexicon: Constructing publicly available language resources for Arabic sentiment analysis. *J. Inf. Sci.* **2018**, *44*, 345–362. [CrossRef]
134. Mohammad, S.; Salameh, M.; Kiritchenko, S. Sentiment Lexicons for Arabic Social Media. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), Portorož, Slovenia, 23–28 May 2016; pp. 33–37.
135. Al-Sallab, A.; Baly, R.; Hajj, H.; Shaban, K.B.; El-Hajj, W.; Badaro, G. AROMA: A Recursive Deep Learning Model for Opinion Mining in Arabic as a Low Resource Language. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* **2017**, *16*, 25. [CrossRef]

136. Shoukry, A.M. Arabic Sentence-Level Sentiment Analysis. Master's Thesis, American University in Cairo, New Cairo, Egypt, 2013.
137. Abdelrahman, S.E.; Mobarz, H.; Farag, I.; Rashwan, M. Arabic Phrase-Level Contextual Polarity Recognition to Enhance Sentiment Arabic Lexical Semantic Database Generation. *Int. J. Adv. Comput. Sci. Appl.* **2014**, *5*. [CrossRef]
138. Al-Dabet, S.; Tedmori, S.; AL-Smadi, M. Enhancing Arabic aspect-based sentiment analysis using deep learning models. *Comput. Speech Lang.* **2021**, *69*, 101224. [CrossRef]
139. Farra, N.; Challita, E.; Abou Assi, R.; Hajj, H. Sentence-level and document-level sentiment mining for Arabic texts. In Proceedings of the 2010 IEEE International Conference on Data Mining Workshops, Sydney, Australia, 13 December 2010; pp. 1114–1119.
140. Alayba, A.M.; Palade, V.; England, M.; Iqbal, R. A Combined CNN and LSTM Model for Arabic Sentiment Analysis. In Proceedings of the Machine Learning and Knowledge Extraction, Hamburg, Germany, 27–30 August 2018; pp. 179–191.
141. El-Beltagy, S.R. NileULex: A Phrase and Word Level Sentiment Lexicon for Egyptian and Modern Standard Arabic. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), Portorož, Slovenia, 23–28 May 2016; pp. 2900–2905.
142. Alharbi, A.I.; Smith, P.; Lee, M. Integrating Character-level and Word-level Representation for Affect in Arabic Tweets. *Data Knowl. Eng.* **2022**, *138*, 101973. [CrossRef]
143. Duwairi, R.M.; Qarqaz, I. Arabic Sentiment Analysis Using Supervised Classification. In Proceedings of the 2014 International Conference on Future Internet of Things and Cloud, Barcelona, Spain, 27–29 August 2014; pp. 579–583.
144. Duwairi, R.; El-Orfali, M. A study of the effects of preprocessing strategies on sentiment analysis for Arabic text. *J. Inf. Sci.* **2014**, *40*, 501–513. [CrossRef]
145. Duwairi, R.M.; Marji, R.; Sha'ban, N.; Rushaidat, S. Sentiment Analysis *in* Arabic tweets. In Proceedings of the 2014 5th International Conference on Information and Communication Systems (ICICS), Irbid, Jordan, 1–3 April 2014; pp. 1–6.
146. Hamouda, A.E.-D.A.; El-taher, F.E. Sentiment analyzer for Arabic comments system. *IJACSA.* **2013**, *4*. [CrossRef]
147. El-Halees, A.M. Arabic text classification using maximum entropy. *IUG J. Nat. Stud.* **2015**, *15*.
148. Alayba, A.M.; Palade, V.; England, M.; Iqbal, R. Arabic language sentiment analysis on health services. In Proceedings of the 2017 1st International Workshop on Arabic Script Analysis and Recognition (ASAR), Nancy, France, 3–5 April 2017; pp. 114–118.
149. Al-Kabi, M.N.; Gigieh, A.H.; Alsmadi, I.M.; Wahsheh, H.A.; Haidar, M.M. Opinion mining and analysis for Arabic language. *IJACSA Int. J. Adv. Comput. Sci. Appl.* **2014**, *5*, 181–195.
150. Duwairi, R.M. Sentiment analysis for dialectical Arabic. In Proceedings of the 2015 6th International Conference on Information and Communication Systems (ICICS), Amman, Jordan, 7–9 April 2015; pp. 166–170.
151. Abdulla, N.A.; Ahmed, N.A.; Shehab, M.A.; Al-Ayyoub, M.; Al-Kabi, M.N.; Al-rifai, S. Towards Improving the Lexicon-Based Approach for Arabic Sentiment Analysis. *Int. J. Inf. Technol. Web Eng.* **2014**, *9*, 55–71. [CrossRef]
152. Bai, J.; Posner, R.; Wang, T.; Yang, C.; Nabavi, S. Applying deep learning in digital breast tomosynthesis for automatic breast cancer detection: A review. *Med. Image Anal.* **2021**, *71*, 102049. [CrossRef]
153. Kusner, M.; Sun, Y.; Kolkin, N.; Weinberger, K. From Word Embeddings To Document Distances. In Proceedings of the 32nd International Conference on Machine Learning, Lille, France, 6–11 July 2015; pp. 957–966.
154. Mikolov, T.; Chen, K.; Corrado, G.; Dean, J. Efficient Estimation of Word Representations in Vector Space. In Proceedings of the International Conference on Learning Representations, Scottsdale, AZ, USA, 2–4 May 2013.
155. Pennington, J.; Socher, R.; Manning, C. GloVe: Global Vectors for Word Representation. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), Doha, Qatar, 25–29 October 2014; pp. 1532–1543.
156. Joulin, A.; Grave, E.; Bojanowski, P.; Mikolov, T. Bag of Tricks for Efficient Text Classification. In Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers. Valencia, Spain, 3–7 April 2017; pp. 427–431.
157. Al Sallab, A.; Hajj, H.; Badaro, G.; Baly, R.; El Hajj, W.; Bashir Shaban, K. Deep Learning Models for Sentiment Analysis in Arabic. In Proceedings of the Second Workshop on Arabic Natural Language Processing, Beijing, China, 26–31 July 2015; pp. 9–17.
158. Dahou, A.; Xiong, S.; Zhou, J.; Haddoud, M.H.; Duan, P. Word Embeddings and Convolutional Neural Network for Arabic Sentiment Classification. In Proceedings of the COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers, Osaka, Japan, 11–17 December 2016; pp. 2418–2427.
159. Alhumoud, S.O.; Al Wazrah, A.A. Arabic sentiment analysis using recurrent neural networks: A review. *Artif. Intell. Rev.* **2022**, *55*, 707–748. [CrossRef]
160. Wazrah, A.A.; Alhumoud, S. Sentiment Analysis Using Stacked Gated Recurrent Unit for Arabic Tweets. *IEEE Access* **2021**, *9*, 137176–137187. [CrossRef]
161. Al-Smadi, M.; Talafha, B.; Al-Ayyoub, M.; Jararweh, Y. Using long short-term memory deep neural networks for aspect-based sentiment analysis of Arabic reviews. *Int. J. Mach. Learn. Cyber.* **2019**, *10*, 2163–2175. [CrossRef]

162. Antoun, W.; Baly, F.; Hajj, H. AraBERT: Transformer-based Model for Arabic Language Understanding. In Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection, Marseille, France, 12 May 2020; pp. 9–15.
163. Mohamed, O.; Kassem, A.M.; Ashraf, A.; Jamal, S.; Mohamed, E.H. An ensemble transformer-based model for Arabic sentiment analysis. *Soc. Netw. Anal. Min.* **2022**, *13*, 11. [CrossRef]
164. Torrey, L.; Shavlik, J. Transfer Learning. In *Handbook of Research on Machine Learning Applications and Trends*; Olivas, E.S., Guerrero, J.D.M., Martinez-Sober, M., Magdalena-Benedito, J.R., Serrano López, A.J., Eds.; IGI Global: Palmdale, PA, USA, 2010; pp. 242–264, ISBN 978-1-60566-766-9.
165. Bensoltane, R.; Zaki, T. Towards Arabic aspect-based sentiment analysis: A transfer learning-based approach. *Soc. Netw. Anal. Min.* **2022**, *12*, 7. [CrossRef]
166. Alayba, A.M.; Palade, V. Leveraging Arabic sentiment classification using an enhanced CNN-LSTM approach and effective Arabic text preparation. *J. King Saud Univ. Comput. Inf. Sci.* **2022**, *34*, 9710–9722. [CrossRef]
167. Alharbi, O. A deep learning approach combining CNN and Bi-LSTM with SVM classifier for Arabic sentiment analysis. *Int. J. Adv. Comput. Sci. Appl.* **2021**, *12*, 165–172. [CrossRef]
168. Aldayel, H.K.; Azmi, A.M. Arabic tweets sentiment analysis—A hybrid scheme. *J. Inf. Sci.* **2016**, *42*, 782–797. [CrossRef]
169. Nabil, M.; Aly, M.; Atiya, A. ASTD: Arabic Sentiment Tweets Dataset. In Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing, Lisbon, Portugal, 17–21 September 2015; pp. 2515–2519.
170. Pontiki, M.; Galanis, D.; Papageorgiou, H.; Androutsopoulos, I.; Manandhar, S.; AL-Smadi, M.; Al-Ayyoub, M.; Zhao, Y.; Qin, B.; De Clercq, O.; et al. SemEval-2016 Task 5: Aspect Based Sentiment Analysis. In Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016), San Diego, CA, USA, 16–17 June 2016; pp. 19–30.
171. Rosenthal, S.; Farra, N.; Nakov, P. SemEval-2017 Task 4: Sentiment Analysis in Twitter. In Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017), Vancouver, BC, Canada, 3–4 August 2017; pp. 502–518.
172. Abdulla, N.A.; Ahmed, N.A.; Shehab, M.A.; Al-Ayyoub, M. Arabic sentiment analysis: Lexicon-based and corpus-based. In Proceedings of the 2013 IEEE Jordan Conference on Applied Electrical Engineering and Computing Technologies (AEECT), Amman, Jordan, 3–5 December 2013; pp. 1–6.
173. Nabil, M.; Aly, M.; Atiya, A. LABR: A Large Scale Arabic Sentiment Analysis Benchmark. *arXiv* **2014**, arXiv:1411.6718. [CrossRef]
174. Elnagar, A.; Khalifa, Y.S.; Einea, A. Hotel Arabic-Reviews Dataset Construction for Sentiment Analysis Applications. In *Intelligent Natural Language Processing: Trends and Applications*; Shaalan, K., Hassanien, A.E., Tolba, F., Eds.; Studies in Computational Intelligence; Springer International Publishing: Cham, Switzerland, 2018; Volume 740, pp. 35–52, ISBN 978-3-319-67055-3.
175. Alayba, A.M.; Palade, V.; England, M.; Iqbal, R. Improving Sentiment Analysis in Arabic Using Word Representation. In Proceedings of the 2018 IEEE 2nd International Workshop on Arabic and Derived Script Analysis and Recognition (ASAR), London, UK, 12–14 March 2018; pp. 13–18.
176. Gamal, D.; Alfonse, M.; El-Horbaty, E.-S.M.; Salem, A.-B.M. Twitter benchmark dataset for Arabic sentiment analysis. *Int. J. Mod. Educ. Comput. Sci.* **2019**, *11*, 33. [CrossRef]
177. Al-Twairesh, N.; Al-Khalifa, H.; Al-Salman, A.; Al-Ohali, Y. AraSenTi-Tweet: A Corpus for Arabic Sentiment Analysis of Saudi Tweets. *Procedia Comput. Sci.* **2017**, *117*, 63–72. [CrossRef]
178. Alnawas, A.; Arıcı, N. The corpus based approach to sentiment analysis in modern standard Arabic and Arabic dialects: A literature review. *Politek. Derg.* **2018**, *21*, 461–470. [CrossRef]
179. Mhamed, M.; Sutcliffe, R.; Quteineh, H.; Sun, X.; Almekhlafi, E.; Retta, E.A.; Feng, J. A deep CNN architecture with novel pooling layer applied to two Sudanese Arabic sentiment data sets. *J. Inf. Sci.* **2023**, 01655515231188341. [CrossRef]
180. Alnawas, A.; Arici, N. Sentiment analysis of Iraqi Arabic dialect on Facebook based on distributed representations of documents. *ACM Trans. Asian Low-Resour. Lang. Inf. Process. (TALLIP)* **2019**, *18*, 20. [CrossRef]
181. Oussous, A.; Benjelloun, F.-Z.; Lahcen, A.A.; Belfkih, S. ASA: A framework for Arabic sentiment analysis. *J. Inf. Sci.* **2020**, *46*, 544–559. [CrossRef]
182. Omran, T.M.; Sharef, B.T.; Grosan, C.; Li, Y. Transfer learning and sentiment analysis of Bahraini dialects sequential text data using multilingual deep learning approach. *Data Knowl. Eng.* **2023**, *143*, 102106. [CrossRef]
183. Abdelli, A.; Guerrouf, F.; Tibermacine, O.; Abdelli, B. Sentiment Analysis of Arabic Algerian Dialect Using a Supervised Method. In Proceedings of the 2019 International Conference on Intelligent Systems and Advanced Computing Sciences (ISACS), Taza, Morocco, 26–27 December 2019; pp. 1–6.
184. Shoukry, A.S.; Rafea, A. A Hybrid Approach for Sentiment Classification of Egyptian Dialect Tweets. In Proceedings of the 2015 First International Conference on Arabic Computational Linguistics (ACLing), Cairo, Egypt, 17–20 April 2015; pp. 78–85.
185. Al Shamsi, A.; Abdallah, S. Sentiment Analysis of Emirati Dialect. *Big Data Cogn. Comput.* **2022**, *6*, 57. [CrossRef]
186. Atoum, J.O.; Nouman, M. Sentiment analysis of Arabic Jordanian dialect tweets. *Int. J. Adv. Comput. Sci. Appl.* **2019**, *10*, 256–262. [CrossRef]

187. Haraty, R.; Chehade, M. Transfer learning and sentiment analysis of lebanese dialect data using a multilingual deep learning approach. *Int. J. Speech Technol.* **2025**, *28*, 581–595. [CrossRef]
188. Zoroub, M.K.; Maghari, A.Y.; Alashqar, A.M. Sentiment Analysis of Palestinian Arabic Dialect Using Lexicon-Based Approach. *Int. J. Comput. Digit. Syst.* **2024**, *16*, 1–10. [CrossRef]
189. Medhaffar, S.; Bougares, F.; Estève, Y.; Hadrich-Belguith, L. Sentiment Analysis of Tunisian Dialects: Linguistic Ressources and Experiments. In Proceedings of the Third Arabic Natural Language Processing Workshop, Valencia, Spain, 03 April 2017; pp. 55–61.
190. Najar, D.; Mesfar, S. Opinion mining and sentiment analysis for Arabic on-line texts: Application on the political domain. *Int. J. Speech Technol.* **2017**, *20*, 575–585. [CrossRef]
191. Almutairi, S.M.; Alotaibi, F.M. A comparative analysis for Arabic sentiment analysis models in e-marketing using deep learning techniques. *J. Eng. Appl. Sci.* **2023**, *10*, 19. [CrossRef]
192. Almaqtari, H.; Zeng, F.; Mohammed, A. Enhancing Arabic Sentiment Analysis of Consumer Reviews: Machine Learning and Deep Learning Methods Based on NLP. *Algorithms* **2024**, *17*, 495. [CrossRef]
193. Alayba, A. Twitter Sentiment Analysis on Health Services in Arabic. Ph.D. Thesis, Coventry University, Coventry, UK, 2019.
194. Alzahrani, M.; AlGhamdi, F. Social Media Sentiment Analysis for Sustainable Rural Event Planning: A Case Study of Agricultural Festivals in Al-Baha, Saudi Arabia. *Sustainability* **2025**, *17*, 3864. [CrossRef]
195. Alomari, E.; Mehmood, R.; Katib, I. Sentiment analysis of Arabic tweets for road traffic congestion and event detection. In *Smart Infrastructure and Applications: Foundations for Smarter Cities and Societies*; Springer: Berlin/Heidelberg, Germany, 2019; pp. 37–54.
196. Basabain, S.; Al-Dubai, A.; Cambria, E.; Alomar, K.; Hussain, A. Arabic Short-Text Dataset for Sentiment Analysis of Tourism and Leisure Events. *Expert Syst.* **2025**, *42*, e70030. [CrossRef]
197. Ishac, W.; Javani, V.; Youssef, D. Leveraging sentiment analysis of Arabic tweets for the 2022 FIFA world cup insights, incorporating the gulf region. *Manag. Sport Leis.* **2024**, 1–17. [CrossRef]
198. Abbas, M.; Smaili, K. Comparison of topic identification methods for arabic language. In Proceedings of the International Conference RANLP-2005 (Recent Advances in Natural Language Processing), Borovets, Bulgaria, 21–23 September 2005; pp. 14–17.
199. Moh'd A Mesleh, A. Chi square feature extraction based svms Arabic text categorization system. In Proceedings of the Second International Conference on Software and Data Technologies-PL/DPS/KE/WsMUSE, Barcelona, Spain, 22–25 July 2007; Volume 2, pp. 235–240.
200. Saad, M.K.; Ashour, W. Osac: Open source Arabic corpora. In Proceedings of the 6th ArchEng International Symposiums, EEECS'10, Lefke, North Cyprus, 25–26 November 2010; Volume 10, p. 55.
201. Abbas, M.; Smaïli, K.; Berkani, D. Evaluation of topic identification methods on Arabic corpora. *J. Digit. Inf. Manag.* **2011**, *9*, 185–192.
202. Selab, E.; Guessoum, A. Building TALAA, a Free General and Categorized Arabic Corpus. In Proceedings of the International Conference on Agents and Artificial Intelligence-Volume 1, Setubal, Portugal, 10–12 January 2015; pp. 284–291.
203. Chouigui, A.; Khiroun, O.B.; Elayeb, B. ANT Corpus: An Arabic News Text Collection for Textual Classification. In Proceedings of the 2017 IEEE/ACS 14th International Conference on Computer Systems and Applications (AICCSA), Hammamet, Tunisia, 30 October–3 November 2017; pp. 135–142.
204. Boukil, S.; Biniz, M.; Adnani, F.E.; Cherrat, L.; Moutaouakkil, A.E.E. Arabic Text Classification Using Deep Learning Technics. *Int. J. Grid Distrib. Comput.* **2018**, *11*, 103–114. [CrossRef]
205. Galal, M.; Madbouly, M.M.; El-Zoghby, A. Classifying Arabic text using deep learning. *J. Theor. Appl. Inf. Technol.* **2019**, *97*, 3412–3422.
206. Einea, O.; Elnagar, A.; Al Debsi, R. SANAD: Single-label Arabic News Articles Dataset for Automatic Text Categorization. *Data Brief* **2019**, *25*, 104076. [CrossRef] [PubMed]
207. Al-Salemi, B.; Ayob, M.; Kendall, G.; Noah, S.A.M. Multi-label Arabic text categorization: A benchmark and baseline comparison of multi-label learning algorithms. *Inf. Process. Manag.* **2019**, *56*, 212–227. [CrossRef]
208. Al-Debsi, R.; Elnagar, A.; Einea, O. *NADiA: News Articles Dataset in Arabic for Multi-Label Text Categorization*; Elsevier: Amsterdam, The Netherlands, 2019. [CrossRef]
209. Almuzaini, H.A.; Azmi, A.M. Impact of Stemming and Word Embedding on Deep Learning-Based Arabic Text Categorization. *IEEE Access* **2020**, *8*, 127913–127928. [CrossRef]
210. Bdeir, A.M.; Ibrahim, F. A Framework for Arabic Tweets Multi-label Classification Using Word Embedding and Neural Networks Algorithms. In Proceedings of the 2020 2nd International Conference on Big Data Engineering, New York, NY, USA, 29–31 May 2020; pp. 105–112.
211. Altamimi, M.; Alayba, A.M. ANAD: Arabic news article dataset. *Data Brief* **2023**, *50*, 109460. [CrossRef] [PubMed]
212. Bouchiha, D.; Bouziane, A.; Doumi, N.; Berbouchi, F.O.; Kebir, A.A.; Mebarki, N.; Benameur, B.A. WiHArD: Wikipedia Based Hierarchical Arabic Dataset for Text Classification. In Proceedings of the 2024 4th International Conference on Embedded & Distributed Systems (EDiS), Bechar, Algeria, 3–5 November 2024; pp. 115–118.

213. Farha, I.A.; Magdy, W. From Arabic Sentiment Analysis to Sarcasm Detection: The ArSarcasm Dataset. In Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, Paris, France, 11–16 May 2020; pp. 32–39.
214. Alayba, A.M.; Altamimi, M. Optimization of Arabic text classification using SVM integrated with word embedding models on a novel dataset. *Int. J. Adv. Appl. Sci.* **2025**, *12*, 140–151. [CrossRef]
215. Alayba, A. *ANACD-Arabic-News-Article-Classification-Dataset*; Elsevier: Amsterdam, The Netherlands, 2025. [CrossRef]
216. Tanfouri, I.; Tlik, G.; Jarray, F. An automatic arabic text summarization system based on genetic algorithms. *Procedia Comput. Sci.* **2021**, *189*, 195–202. [CrossRef]
217. Jaafar, Y.; Bouzoubaa, K. Towards a New Hybrid Approach for Abstractive Summarization. *Procedia Comput. Sci.* **2018**, *142*, 286–293. [CrossRef]
218. Giarelis, N.; Mastrokostas, C.; Karacapilidis, N. Abstractive vs. Extractive Summarization: An Experimental Review. *Appl. Sci.* **2023**, *13*, 7620. [CrossRef]
219. Omar, K.; Al-Shaar, M. Method for Arabic text Summarization using statistical features and word2vector approach. In Proceedings of the 2023 9th International Conference on Computer Technology Applications, Vienna, Austria, 20 August 2023; pp. 258–262.
220. AL-Khassawneh, Y.A.; Hanandeh, E.S. Extractive Arabic Text Summarization-Graph-Based Approach. *Electronics* **2023**, *12*, 437. [CrossRef]
221. Wazery, Y.M.; Saleh, M.E.; Alharbi, A.; Ali, A.A. Abstractive Arabic Text Summarization Based on Deep Learning. *Comput. Intell. Neurosci.* **2022**, *2022*, 1566890. [CrossRef]
222. Etaiwi, W.; Awajan, A. SemG-TS: Abstractive Arabic Text Summarization Using Semantic Graph Embedding. *Mathematics* **2022**, *10*, 3225. [CrossRef]
223. Al Qassem, L.; Wang, D.; Barada, H.; Al-Rubaie, A.; Almoosa, N. Automatic Arabic text summarization based on fuzzy logic. In Proceedings of the 3rd International Conference on Natural Language and Speech Processing, Trento, Italy, 12–13 September 2019; pp. 42–48.
224. Belkebir, R.; Guessoum, A. A Supervised Approach to Arabic Text Summarization Using AdaBoost. In *New Contributions in Information Systems and Technologies*; Rocha, A., Correia, A.M., Costanzo, S., Reis, L.P., Eds.; Advances in Intelligent Systems and Computing; Springer International Publishing: Cham, Switzerland, 2015; Volume 353, pp. 227–236, ISBN 978-3-319-16485-4.
225. Alshemaimri, B.; Alrayes, I.; Alothman, T.; Almalik, F.; Almotlaq, M. Summarizing Arabic Articles using Large Language Models. In Proceedings of the Advanced Natural Language Processing, May 2024.
226. Bani-Almarjeh, M.; Kurdy, M.-B. Arabic abstractive text summarization using RNN-based and transformer-based architectures. *Inf. Process. Manag.* **2023**, *60*, 103227. [CrossRef]
227. R Reda, A.; Salah, N.; Adel, J.; Ehab, M.; Ahmed, I.; Magdy, M.; Khoriba, G.; Mohamed, E.H. A Hybrid Arabic Text Summarization Approach based on Transformers. In Proceedings of the 2022 2nd International Mobile, Intelligent, and Ubiquitous Computing Conference (MIUCC), Cairo, Egypt, 8–9 May 2022; pp. 56–62.
228. El-Haj, M.; Kruschwitz, U.; Fox, C. Creating language resources for under-resourced languages: Methodologies, and experiments with Arabic. *Lang. Resour. Eval.* **2015**, *49*, 549–580. [CrossRef]
229. Alhamadani, A.; Zhang, X.; He, J.; Khatri, A.; Lu, C.-T. LANS: Large-scale Arabic News Summarization Corpus. In Proceedings of the ArabicNLP 2023, Singapore (hybrid conference), 7 December 2023; pp. 89–100.
230. Almarjeh, M.B. *SumArabic*; Elsevier: Amsterdam, The Netherlands, 2022. [CrossRef]
231. Kahla, M.; Yang, Z.G.; Novák, A. Cross-lingual Fine-tuning for Abstractive Arabic Text Summarization. In Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2021), Online, 1–3 September 2021; pp. 655–663.
232. Al-Maleh, M.; Desouki, S. Arabic text summarization using deep learning approach. *J. Big Data* **2020**, *7*, 109. [CrossRef]
233. Pu, X.; Gao, M.; Wan, X. Is Summary Useful or Not? An Extrinsic Human Evaluation of Text Summaries on Downstream Tasks. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), Torino, Italy, 20–25 May 2024; pp. 9389–9404.
234. Lin, C.-Y. ROUGE: A Package for Automatic Evaluation of Summaries. In Proceedings of the Text Summarization Branches Out, Barcelona, Spain, 25–26 July 2004; pp. 74–81.
235. Alahmadi, D.; Wali, A.; Alzahrani, S. TAAM: Topic-aware abstractive Arabic text summarisation using deep recurrent neural networks. *J. King Saud Univ. Comput. Inf. Sci.* **2022**, *34*, 2651–2665. [CrossRef]
236. Al-Numai, A.; Azmi, A. LEMMA-ROUGE: An Evaluation Metric for Arabic Abstractive Text Summarization. *Indones. J. Comput. Sci.* **2023**, *12*, 470–481. [CrossRef]
237. Al-Khawaldeh, F.; Samawi, V. Lexical cohesion and entailment based segmentation for Arabic text summarization (lceas). *World Comput. Sci. Inf. Technol. J. (WSCIT)* **2015**, *5*, 51–60.
238. Azmi, A.M.; Al-Thanyyan, S. A text summarizer for Arabic. *Comput. Speech Lang.* **2012**, *26*, 260–273. [CrossRef]

239. Elayeb, B.; Chouigui, A.; Bounhas, M.; Khiroun, O.B. Automatic Arabic Text Summarization Using Analogical Proportions. *Cogn. Comput.* **2020**, *12*, 1043–1069. [CrossRef]
240. Hammo, B.; Abu-Salem, H.; Lytinen, S. QARAB: A question answering system to support the Arabic language. In Proceedings of the ACL-02 workshop on Computational Approaches to Semitic Languages, Philadelphia, PA, USA, July 2002; pp. 1–11.
241. Maraoui, H.; Haddar, K.; Romary, L. Arabic factoid Question-Answering system for Islamic sciences using normalized corpora. *Procedia Comput. Sci.* **2021**, *192*, 69–79. [CrossRef]
242. Saadaoui, Z.; Tlig, G.; Jarray, F. LLMs Based Approach for Quranic Question Answering. 2024. Available online: https://www.scitepress.org/Papers/2024/130129/130129.pdf (accessed on 17 September 2025).
243. Al-Smadi, M.; Al-Dalabih, I.; Jararweh, Y.; Juola, P. Leveraging Linked Open Data to Automatically Answer Arabic Questions. *IEEE Access* **2019**, *7*, 177122–177136. [CrossRef]
244. Bdour, W.N.; Gharaibeh, N.K. Development of Yes/No Arabic Question Answering System. *Int. J. Artif. Intell. Appl.* **2013**, *4*, 51–63. [CrossRef]
245. Azmi, A.M.; Alshenaifi, N.A. Answering Arabic Why-Questions: Baseline vs. RST-Based Approach. *ACM Trans. Inf. Syst.* **2017**, *35*, 6. [CrossRef]
246. Mozannar, H.; Maamary, E.; El Hajal, K.; Hajj, H. Neural Arabic Question Answering. In Proceedings of the Fourth Arabic Natural Language Processing Workshop, Florence, Italy, 1 August 2019; pp. 108–118.
247. Ismail, W.S.; Homsi, M.N. DAWQAS: A Dataset for Arabic Why Question Answering System. *Procedia Comput. Sci.* **2018**, *142*, 123–131. [CrossRef]
248. Aleid, H.A.; Azmi, A.M. Hajj-FQA: A benchmark Arabic dataset for developing question-answering systems on Hajj fatwas. *J. King Saud Univ. Comput. Inf. Sci.* **2025**, *37*, 135. [CrossRef]
249. Abouenour, L.; Bouzouba, K.; Rosso, P. An evaluated semantic query expansion and structure-based approach for enhancing Arabic question/answering. *Int. J. Inf. Commun. Technol.* **2010**, *3*, 37–51.
250. Atef, A.; Mattar, B.; Sherif, S.; Elrefai, E.; Torki, M. AQAD: 17,000+ Arabic Questions for Machine Comprehension of Text. In Proceedings of the 2020 IEEE/ACS 17th International Conference on Computer Systems and Applications (AICCSA), Antalya, Turkey, 2–5 November 2020; pp. 1–6.
251. Alami, H.; El Mahdaouy, A.; Benlahbib, A.; En-Nahnahi, N.; Berrada, I.; Ouatik, S.E.A. DAQAS: Deep Arabic Question Answering System based on duplicate question detection and machine reading comprehension. *J. King Saud Univ. Comput. Inf. Sci.* **2023**, *35*, 101709. [CrossRef]
252. Kamel, S.M.; Hassan, S.I.; Elrefaei, L. VAQA: Visual Arabic Question Answering. *Arab. J. Sci. Eng.* **2023**, *48*, 10803–10823. [CrossRef]
253. Hadla, L.S.; Hailat, T.M.; Al-Kabi, M.N. Evaluating Arabic to English machine translation. *Int. J. Adv. Comput. Sci. Appl.* **2014**, *5*, 68–73. [CrossRef]
254. Harrat, S.; Meftouh, K.; Smaili, K. Machine translation for Arabic dialects (survey). *Inf. Process. Manag.* **2019**, *56*, 262–273. [CrossRef]
255. Elfardy, H.; Al-Badrashiny, M.; Diab, M. AIDA: Identifying Code Switching in Informal Arabic Text. In Proceedings of the First Workshop on Computational Approaches to Code Switching, Doha, Qatar, 25 October 2014; pp. 94–101.
256. Husin, M.Z.; Saad, S.; Noah, S.A.M. Syntactic rule-based approach for extracting concepts from quranic translation text. In Proceedings of the 2017 6th International Conference on Electrical Engineering and Informatics (ICEEI), Langkawi, Malaysia, 25–27 November 2017; pp. 1–6.
257. Hatem, A.; Omar, N.; Shaker, K. Morphological analysis for rule based machine translation. In Proceedings of the 2011 International Conference on Semantic Technology and Information Retrieval, Putrajaya, Malaysia, 28–29 June 2011; pp. 260–263.
258. Alqudsi, A.; Omar, N.; Shaker, K. Arabic machine translation: A survey. *Artif. Intell. Rev.* **2014**, *42*, 549–572. [CrossRef]
259. Badr, I.; Zbib, R.; Glass, J. Segmentation for English-to-Arabic Statistical Machine Translation. In Proceedings of the ACL-08: HLT, Short Papers. Columbus, OH, USA, 16–17 June 2008; pp. 153–156.
260. Shapiro, P.; Duh, K. Morphological Word Embeddings for Arabic Neural Machine Translation in Low-Resource Settings. In Proceedings of the Second Workshop on Subword/Character LEvel Models, New Orleans, LA, USA, 6 June 2018; pp. 1–11.
261. Bari, M.S.; Alnumay, Y.; Alzahrani, N.A.; Alotaibi, N.M.; Alyahya, H.A.; AlRashed, S.; Mirza, F.A.; Alsubaie, S.Z.; Alahmed, H.A.; Alabduljabbar, G.; et al. ALLaM: Large Language Models for Arabic and English. *arXiv* **2024**, arXiv:2407.15390. [CrossRef]
262. Abdelaziz, A.A.A.; Elneima, A.H.; Darwish, K. LLM-based MT Data Creation: Dialectal to MSA Translation Shared Task. In Proceedings of the 6th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT) with Shared Tasks on Arabic LLMs Hallucination and Dialect to MSA Machine Translation @ LREC-COLING 2024, Torino, Italy, 20–25 May 2024; pp. 112–116.
263. Alqudsi, A.; Omar, N.; Shaker, K. A Hybrid Rules and Statistical Method for Arabic to English Machine Translation. In Proceedings of the 2019 2nd International Conference on Computer Applications & Information Security (ICCAIS), Riyadh, Saudi Arabia, 1–3 May 2019; pp. 1–7.

264. Ziemski, M.; Junczys-Dowmunt, M.; Pouliquen, B. The United Nations Parallel Corpus v1. In 0. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), Portorož, Slovenia, 23–28 May 2016; pp. 3530–3534.
265. Zaghouani, W.; Habash, N.; Mohit, B. *The Qatar Arabic Language Bank Guidelines*; Technical Report CMU-CS-QTR-124; School of Computer Science, Carnegie Mellon University: Pittsburgh, PA, USA, 2014.
266. Bouamor, H.; Habash, N.; Salameh, M.; Zaghouani, W.; Rambow, O.; Abdulrahim, D.; Obeid, O.; Khalifa, S.; Eryani, F.; Erdmann, A.; et al. The MADAR Arabic Dialect Corpus and Lexicon. In Proceedings of the International Conference on Language Resources and Evaluation, Miyazaki, Japan, 7–12 May 2018.
267. Alzamzami, F.; Saddik, A.E. OSN-MDAD: Machine Translation Dataset for Arabic Multi-Dialectal Conversations on Online Social Media. *arXiv* **2023**, arXiv:2309.12137. [CrossRef]
268. Khered, A.; Benkhedda, Y.; Batista-Navarro, R. Dial2MSA-Verified: A Multi-Dialect Arabic Social Media Dataset for Neural Machine Translation to Modern Standard Arabic. In Proceedings of the 4th Workshop on Arabic Corpus Linguistics (WACL-4), Abu Dhabi, United Arab Emirates, 20 January 2025; pp. 50–62.
269. Shirko, O.; Omar, N.; Arshad, H.; Albared, M. Machine translation of noun phrases from Arabic to English using transfer-based approach. *J. Comput. Sci.* **2010**, *6*, 350. [CrossRef]
270. Almahasees, Z.M. Assessment of Google and Microsoft Bing Translation of Journalistic Texts. *Int. J. Lang. Lit. Linguist.* **2018**, *4*, 231–235. [CrossRef]
271. Bouamor, H.; Alshikhabobakr, H.; Mohit, B.; Oflazer, K. A Human Judgement Corpus and a Metric for Arabic MT Evaluation. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), Doha, Qatar, 25–29 October 2014; pp. 207–213.
272. Condon, S.; Parvaz, D.; Aberdeen, J.; Doran, C.; Freeman, A.; Awad, M. *Evaluation of Machine Translation Errors in English and Iraqi Arabic*; Defense Technical Information Center: Fort Belvoir, VA, USA, 2010.
273. Al Amer, S.A.; Lee, M.G.; Smith, P. Comparative Evaluation of Machine Translation Models Using Human-Translated Social Media Posts as References: Human-Translated Datasets. In Proceedings of the Eighth Workshop on Technologies for Machine Translation of Low-Resource Languages (LoResMT 2025), Albuquerque, NM, USA, 3–4 May 2025; pp. 1–9.
274. Alabdullah, A.; Han, L.; Lin, C. Advancing Dialectal Arabic to Modern Standard Arabic Machine Translation. *arXiv* **2025**, arXiv:2507.20301. [CrossRef]
275. Devlin, J.; Chang, M.-W.; Lee, K.; Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers, Minneapolis, MN, USA, 2–7 June 2019; pp. 4171–4186.
276. Abdul-Mageed, M.; Elmadany, A.; Nagoudi, E.M.B. ARBERT & MARBERT: Deep Bidirectional Transformers for Arabic. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), Online, 1–6 August 2021; pp. 7088–7105.
277. Inoue, G.; Alhafni, B.; Baimukan, N.; Bouamor, H.; Habash, N. The Interplay of Variant, Size, and Task Type in Arabic Pre-trained Language Models. In Proceedings of the Sixth Arabic Natural Language Processing Workshop, Kyiv, Ukraine, 19 April 2021; pp. 92–104.
278. Ray, P.P. ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope. *Internet Things Cyber-Phys. Syst.* **2023**, *3*, 121–154. [CrossRef]
279. Antoun, W.; Baly, F.; Hajj, H. AraGPT2: Pre-Trained Transformer for Arabic Language Generation. In Proceedings of the Sixth Arabic Natural Language Processing Workshop, Kyiv, Ukraine, 19 April 2021; pp. 196–207.
280. Koubaa, A.; Ammar, A.; Ghouti, L.; Najar, O.; Sibaee, S. ArabianGPT: Native Arabic GPT-based Large Language Model. *arXiv* **2024**, arXiv:2402.15313. [CrossRef]
281. Raffel, C.; Shazeer, N.; Roberts, A.; Lee, K.; Narang, S.; Matena, M.; Zhou, Y.; Li, W.; Liu, P.J. Exploring the limits of transfer learning with a unified text-to-text transformer. *J. Mach. Learn. Res.* **2020**, *21*, 5485–5551.
282. Nagoudi, E.M.B.; Elmadany, A.; Abdul-Mageed, M. AraT5: Text-to-Text Transformers for Arabic Language Generation. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Dublin, Ireland, 22–27 May 2022; pp. 628–647.
283. Alghamdi, A.; Duan, X.; Jiang, W.; Wang, Z.; Wu, Y.; Xia, Q.; Wang, Z.; Zheng, Y.; Rezagholizadeh, M.; Huai, B.; et al. AraMUS: Pushing the Limits of Data and Model Scale for Arabic Natural Language Processing. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2023, Toronto, ON, Canada, 9–14 July 2023; pp. 2883–2894.
284. Qachfar, F.Z.; Verma, R. ReDASPersuasion at ArAIEval Shared Task: Multilingual and Monolingual Models For Arabic Persuasion Detection. In Proceedings of the ArabicNLP 2023, Singapore (hybrid conference), 7 December 2023; pp. 549–557.
285. Billah Nagoudi, E.M.; Abdul-Mageed, M.; Elmadany, A.; Inciarte, A.; Islam Khondaker, M.T. JASMINE: Arabic GPT Models for Few-Shot Learning. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, Singapore, 6–10 December 2023; pp. 16721–16744.

286. AlYami, R.; Al-Zaidy, R. Weakly and Semi-Supervised Learning for Arabic Text Classification using Monodialectal Language Models. In Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP), Abu Dhabi, United Arab Emirates, 8 December 2022; pp. 260–272.
287. Qarah, F. SaudiBERT: A Large Language Model Pretrained on Saudi Dialect Corpora. *arXiv* **2024**, arXiv:2405.06239. [CrossRef]
288. Elgezouli, M.; Elmadani, K.N.; Saeed, M. SudaBERT: A Pre-trained Encoder Representation For Sudanese Arabic Dialect. In Proceedings of the 2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE), Khartoum, Sudan, 26 February–1 March 2021; pp. 1–4.
289. Abdaoui, A.; Berrimi, M.; Oussalah, M.; Moussaoui, A. DziriBERT: A Pre-trained Language Model for the Algerian Dialect. *arXiv* **2021**, arXiv:2109.12346. [CrossRef]
290. Moussaoui, O.; El Younnoussi, Y. Pre-training Two BERT-Like Models for Moroccan Dialect: MorRoBERTa and MorrBERT. *Mendel* **2023**, *29*, 55–61. [CrossRef]
291. Gaanoun, K.; Naira, A.M.; Allak, A.; Benelallam, I. DarijaBERT: A step forward in NLP for the written Moroccan dialect. *Int. J. Data Sci. Anal.* **2025**, *20*, 917–929. [CrossRef]
292. Shang, G.; Abdine, H.; Khoubrane, Y.; Mohamed, A.; Abbahaddou, Y.; Ennadir, S.; Momayiz, I.; Ren, X.; Moulines, E.; Nakov, P.; et al. Atlas-Chat: Adapting Large Language Models for Low-Resource Moroccan Arabic Dialect. In Proceedings of the First Workshop on Language Models for Low-Resource Languages, Abu Dhabi, United Arab Emirates, 20 January 2025; pp. 9–30.
293. Haddad, H.; Rouhou, A.C.; Messaoudi, A.; Korched, A.; Fourati, C.; Sellami, A.; Ben HajHmida, M.; Ghriss, F. TunBERT: Pre-training BERT for Tunisian Dialect Understanding. *SN Comput. Sci.* **2023**, *4*, 194. [CrossRef]
294. Qarah, F. EgyBERT: A Large Language Model Pretrained on Egyptian Dialect Corpora. *arXiv* **2024**, arXiv:2408.03524. [CrossRef]
295. Ahmed, M.; Alfasly, S.; Wen, B.; Addeen, J.; Ahmed, M.; Liu, Y. AlclaM: Arabic Dialect Language Model. In Proceedings of the Second Arabic Natural Language Processing Conference, Bangkok, Thailand, 16 August 2024; pp. 153–159.
296. Alkaoud, M. A bilingual benchmark for evaluating large language models. *PeerJ Comput. Sci.* **2024**, *10*, e1893. [CrossRef]
297. Lan, W.; Chen, Y.; Xu, W.; Ritter, A. An Empirical Study of Pre-trained Transformers for Arabic Information Extraction. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Online, 16–20 November 2020; pp. 4727–4734.
298. Sengupta, N.; Sahu, S.K.; Jia, B.; Katipomu, S.; Li, H.; Koto, F.; Marshall, W.; Gosal, G.; Liu, C.; Chen, Z.; et al. Jais and Jais-chat: Arabic-Centric Foundation and Instruction-Tuned Open Generative Large Language Models. *arXiv* **2023**, arXiv:2308.16149. [CrossRef]
299. Huang, H.; Yu, F.; Zhu, J.; Sun, X.; Cheng, H.; Dingjie, S.; Chen, Z.; Alharthi, M.; An, B.; He, J.; et al. AceGPT, Localizing Large Language Models in Arabic. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), Mexico City, Mexico, 16–21 June 2024; pp. 8139–8163.
300. Alwajih, F.; Nagoudi, E.M.B.; Bhatia, G.; Mohamed, A.; Abdul-Mageed, M. Peacock: A Family of Arabic Multimodal Large Language Models and Benchmarks. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Bangkok, Thailand, 11–16 August 2024; pp. 12753–12776.
301. Abbas, U.; Ahmad, M.S.; Alam, F.; Altinisik, E.; Asgari, E.; Boshmaf, Y.; Boughorbel, S.; Chawla, S.; Chowdhury, S.; Dalvi, F.; et al. Fanar: An Arabic-Centric Multimodal Generative AI Platform. *arXiv* **2025**, arXiv:2501.13944. [CrossRef]
302. Bourahouat, G.; Abourezq, M.; Daoudi, N. Toward an efficient extractive Arabic text summarisation system based on Arabic large language models. *Int. J. Data Sci. Anal.* **2025**, *20*, 2445–2457. [CrossRef]
303. Abdul-Mageed, M.; Keleg, A.; Elmadany, A.; Zhang, C.; Hamed, I.; Magdy, W.; Bouamor, H.; Habash, N. NADI 2024: The Fifth Nuanced Arabic Dialect Identification Shared Task. In Proceedings of the Second Arabic Natural Language Processing Conference, Bangkok, Thailand, 16 August 2024; pp. 709–728.
304. Robinson, N.R.; Abdelmoneim, S.; Marchisio, K.; Ruder, S. AL-QASIDA: Analyzing LLM Quality and Accuracy Systematically in Dialectal Arabic. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2025, Vienna, Austria, 27 July–1 August 2025; pp. 22048–22065.
305. Al-Matham, R.; Darwish, K.; Al-Rasheed, R.; Alshammari, W.; Alhoshan, M.; Almazrua, A.; Wazrah, A.A.; Alheraki, M.; Alam, F.; Nakov, P.; et al. BALSAM: A Platform for Benchmarking Arabic Large Language Models. *arXiv* **2025**, arXiv:2507.22603. [CrossRef]
306. Ashraf, Y.; Wang, Y.; Gu, B.; Nakov, P.; Baldwin, T. Arabic Dataset for LLM Safeguard Evaluation. In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), Albuquerque, NM, USA, 29 April–4 May 2025; pp. 5529–5546.

307. Nacar, O.; Sibaee, S.T.; Ahmed, S.; Ben Atitallah, S.; Ammar, A.; Alhabashi, Y.; Al-Batati, A.S.; Alsehibani, A.; Qandos, N.; Elshehy, O.; et al. Towards Inclusive Arabic LLMs: A Culturally Aligned Benchmark in Arabic Large Language Model Evaluation. In Proceedings of the First Workshop on Language Models for Low-Resource Languages, Abu Dhabi, United Arab Emirates, 20 January 2025; pp. 387–401.
308. Alghamdi, E.A.; Masoud, R.; Alnuhait, D.; Alomairi, A.Y.; Ashraf, A.; Zaytoon, M. AraTrust: An Evaluation of Trustworthiness for LLMs in Arabic. In Proceedings of the 31st International Conference on Computational Linguistics, Abu Dhabi, United Arab Emirates, 19–24 January 2025; pp. 8664–8679.

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